Infinitely Many Primes $p = m^2 + n^2$ with $n < p^{0.1129}$

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October 9, 2026

Abstract

Harman and Lewis proved that infinitely many primes $p = m^2 + n^2$ have $n < p^{0.119}$, and hence that infinitely many primes satisfy $\{\sqrt{p}\} < p^{-0.262}$. Both results follow from a lower bound for the number of Gaussian primes of norm between $X/2$ and X in a sector of angle $X^{\theta-1/2}$. Harman and Lewis prove this bound with Harman's sieve at $\theta = 0.119$. We keep their arithmetic information and add three groupings of terms that their sieve discards. In a two-dimensional sum, we apply Buchstab's identity and then the linear estimate to a factorization that Harman and Lewis do not use. In four-dimensional sums, we apply the bilinear estimate to products of five factors. On another part of these sums, we apply one further Buchstab step, followed by a linear and a bilinear estimate. This gives $\theta = 0.1129$. Consequently, infinitely many primes $p = m^2 + n^2$ have $n < p^{0.1129}$, and infinitely many primes satisfy $\{\sqrt{p}\} < p^{-0.2742}$. The argument of Harman and Lewis leaves several steps unwritten, among them the removal of conditions that couple the variables of their bilinear estimate, the proofs of three sieve lemmas, and a step in one case of their decomposition. We prove every sieve step, correct several statements and loss regions of the original argument, and evaluate the losses by two independent numerical methods.

Keywords: Gaussian primes, narrow sectors, Harman's sieve, Hecke characters, fractional part of $\sqrt{p}$.
2020 MSC: 11N36, 11N32, 11R44, 11J71.

1 Introduction

A prime $p \equiv 1 \pmod 4$ is the norm $p = m^2 + n^2$ of a Gaussian prime $m + in$. Landau asked whether $n = 1$ occurs for infinitely many primes, that is, whether infinitely many primes have the form $m^2 + 1$. The question is open. A weaker problem asks how small n can be taken for infinitely many such primes. With $m > n > 0$, the condition $n < p^\theta$ places the Gaussian prime $m + in$ inside a sector of angle about $p^{\theta-1/2}$ next to the real axis. So the problem is one of Gaussian primes in narrow sectors.

Under the Riemann hypothesis for the Hecke L-functions of $\mathbb{Q}(i)$, Ankeny [1] and Kubilius [8] showed that infinitely many primes $p = m^2 + n^2$ have $n \ll \log p$. Without hypotheses, Coleman [5] obtained $n < p^{0.1631}$ with the Rosser–Iwaniec sieve in number fields, and Harman and Lewis [7] obtained $n < p^{0.119}$ with Harman's sieve.

If $p = m^2 + n^2$ with $n < p^\theta$, then

$$0 < \sqrt{p} - m = \frac{n^2}{\sqrt{p} + m} < p^{2\theta-1/2}. \tag{1}$$

Let $\{x\}$ denote the fractional part of x. Then $\{\sqrt{p}\} < p^{-(1/2-2\theta)}$ once $\theta < 1/4$. From $\theta = 0.119$, Harman and Lewis obtained $\{\sqrt{p}\} < p^{-0.262}$ for infinitely many primes. Earlier, Balog [3] and Harman [6] had obtained the exponent $1/4 - \varepsilon$ for every fixed $\varepsilon > 0$. No exponent above $1/2$ is possible in such a bound, since $|\sqrt{p} - k| \geqslant 1/(2\sqrt{p} + 1)$ for the integer k nearest to $\sqrt{p}$.

Our main result is the following lower bound. Here and below, a non-zero Gaussian integer is identified with its associate of argument in $[0, \pi/2)$, arguments are taken modulo $\pi/2$, and the norm of a Gaussian integer ϖ is $|\varpi|^2$.

Theorem 1 *Let $\theta = 0.1129$. There is a constant $c > 0$ such that, for every $\psi \in \mathbb{R}$ and every sufficiently large X, the number of Gaussian primes ϖ with $X/2 \leqslant |\varpi|^2 < X$ and $\psi \leqslant \arg \varpi < \psi + X^{\theta-1/2}$ is at least $cX^{1/2+\theta}/\log X$.*

Theorem 2 *There are infinitely many primes $p = m^2 + n^2$ with m, n positive integers and $n < p^{0.1129}$. Consequently, there are infinitely many primes p with $\{\sqrt{p}\} < p^{-0.2742}$.*

Proof of Theorem 2 from Theorem 1. Let $\theta = 0.1129$, and apply Theorem 1 with $\psi = 0$ and $X = 2^k$ for large k. The Gaussian primes of argument 0 are the rational primes $q \equiv 3 \pmod 4$, and fewer than $X^{1/2}$ of them have norm $q^2 < X$. So the sector contains a Gaussian prime $\varpi = m + in$ with $m, n > 0$. Its norm $p = m^2 + n^2$ is a rational prime with $X/2 \leqslant p < X$. Since $\theta < 1/2$ and $p < X$,

$$n = \sqrt{p}\,\sin(\arg \varpi) < \sqrt{p}\,X^{\theta-1/2} < \sqrt{p}\,p^{\theta-1/2} = p^{\theta}.$$

The ranges $[X/2, X)$ for different k are disjoint. So the primes p are distinct for different k, and there are infinitely many of them. The second statement follows from (1), because $1/2 - 2\theta = 0.2742$. □

The proof of Theorem 1 refines the argument of Harman and Lewis [7]. Let $\mathcal{A}$ be the set of Gaussian integers in the sector of Theorem 1 with norm in $[X/2, X)$, and let $\mathcal{B}$ be the set of all Gaussian integers of the quadrant with norm in $[X/2, X)$. The number of Gaussian primes in $\mathcal{A}$ is the number of elements of $\mathcal{A}$ without prime factors of norm below $X^{1/2}$. Buchstab's identity (5) decomposes this count into sums of two kinds. Sums of the first kind have an asymptotic formula: they equal $2X^{\theta-1/2}/\pi$ times the corresponding sums for $\mathcal{B}$, up to a negligible error. Harman and Lewis derive these formulas from bounds for Hecke L-functions, through linear (Type I) and bilinear (Type II) estimates. Sums of the second kind consist of non-negative terms that enter with the sign +. They are discarded. Each discarded sum, taken over $\mathcal{B}$ and divided by the number of primes in $\mathcal{B}$, tends to an integral (13) of Buchstab's function ω over a region of exponents. This integral is the loss of the sum, and the sum of the losses is the total loss $\Phi(\theta)$. The result is the lower bound

$$\#\{\varpi \in \mathcal{A} : \varpi \text{ prime}\} \geqslant \frac{2X^{\theta-1/2}}{\pi}\,\#\{\varpi \in \mathcal{B} : \varpi \text{ prime}\}\,\big(1 - \Phi(\theta) - o(1)\big). \tag{2}$$

For their decomposition, Harman and Lewis show that the total loss at $\theta = 0.119$ is below 0.943.

We add three groupings of terms that Harman and Lewis discard. The first acts on a two-dimensional region R of terms pq, with p and q Gaussian primes. There we apply Buchstab's identity, which introduces a third prime r. We then apply the linear estimate to the factorization $p \cdot (qr)$. Harman and Lewis apply the linear estimate after this identity only to the factorization $(pq) \cdot r$. The other two groupings act on four-dimensional regions of terms $pqrs$ with a cofactor t. On a region G of these terms, we apply the bilinear estimate directly to the products of the five factors p, q, r, s, t. On a region L, we apply one further Buchstab step, followed by a linear estimate for $pqrs$ and the bilinear estimate with the new prime. Section 3 defines R, G and L and places them in the decomposition. With these groupings, the total loss equals 1 at $\theta = 0.1128554$ and is below 1 at $\theta = 0.1129$. The loss tables are in Section 10. Without these groupings, and with the regions derived from the lemmas of Section 6 (Remark 27), the decomposition of Harman and Lewis has total loss 1 at $\theta = 0.1167825$. Harman and Lewis state that the constant in their lower bound ceases to be positive between 0.117 and 0.118.

The argument of Harman and Lewis [7] leaves several steps unwritten, and some of its statements need correction. Their bilinear estimate requires coefficients that depend on one variable each. Its applications carry conditions that couple the variables. Harman and Lewis remove these conditions only by references to Perron's formula. Their proofs of the sieve Lemmas 8, 9 and 10 are sketches. In one case of their decomposition, a step is used without a statement. We prove every sieve step. Appendix E lists the departures from the original argument and where each is treated. Besides the prime number theorem in progressions modulo 4, only the arithmetic information is quoted without proof. It consists of the sector inequality, the mean-value estimate for Hecke characters, the prime number theorem with Hecke characters, and the two linear estimates of Harman and Lewis. Lemmas 20 to 23 of Appendix A state them.

Section 2 fixes the notation. Section 3 states the decomposition, the treatment of each of its parts and the total loss, and it names the lemma that proves each step. Section 4 states the linear and bilinear estimates,

a fundamental lemma over $\mathbb{Z}[i]$, and asymptotic formulas for sifting functions with a variable level. Section 5 removes the conditions that couple the variables of a sum. Section 6 treats the sums over two primes, Section 7 the decomposition and its discarded regions, and Section 8 the four-dimensional sums. Section 9 turns the discarded sums into integrals and proves the lower bound, Lemma 16. Section 10 evaluates the integrals by two independent numerical methods and proves Theorem 1. Appendix A collects standard facts on Gaussian primes and derives the linear and bilinear estimates from the arithmetic information. Appendix B proves Lemmas 3, 4, 5 and 7, and Appendix C proves Lemma 15. Appendix D lists the inequalities between the parameters, and Appendix E compares the argument with that of Harman and Lewis.

2 Notation

Throughout, $1/10 < \theta < 1/8$, and X is large in terms of θ and of the other fixed parameters. Following Harman and Lewis [7, p. 120], put

$$\sigma = \frac{1}{2} - \theta, \quad \tau = \frac{1}{2} + \theta, \quad \rho = \frac{1+22\theta}{12}, \quad \zeta = \frac{1+4\theta}{3}, \quad \nu = \frac{7+10\theta}{12}, \tag{3}$$

which satisfy $\nu - \sigma = \rho$ and $\nu - \sigma/2 = \zeta$. Put $W = X^{\sigma}$ and $\mu = \exp((\log X)^{4/5})$. Then $W^{-1} = X^{\theta-1/2}$ is the angle of the sector in Theorem 1, and μ exceeds every fixed power of $\log X$ and is smaller than every fixed positive power of X.

A product of Gaussian integers is identified with the associate of argument in $[0, \pi/2)$ of the product. For a Gaussian integer n, $n \sim N$ means $N \leqslant |n|^2 < 2N$. From now on, p, q, r, s and ϖ denote Gaussian primes, and $d(n)$ is the number of divisors of n. A sequence a_n is *divisor-bounded* if $|a_n| \leqslant d(n)^{C_0}$ for a fixed C_0. Fix $\psi \in \mathbb{R}$ and put

$$\mathcal{A} = \{n : X/2 \leqslant |n|^2 < X,\ \psi \leqslant \arg n < \psi + W^{-1}\}, \qquad \mathcal{B} = \{n : X/2 \leqslant |n|^2 < X\},$$

and write $\mathcal{E}_u = \{n \in \mathcal{E} : u \mid n\}$ for $\mathcal{E} \in \{\mathcal{A}, \mathcal{B}\}$.

Let $\pi_G(y)$ be the number of Gaussian primes of norm less than y. A rational prime $\ell \equiv 1 \pmod 4$ is the norm of 2 Gaussian primes. As in Harman and Lewis [7, p. 122], Gaussian primes are ordered by norm, and primes of equal norm by argument: $\varpi \prec \varpi'$ means $|\varpi| < |\varpi'|$, or $|\varpi| = |\varpi'|$ and $\arg \varpi < \arg \varpi'$. For $z \geqslant 1$ and a prime q, let $S(\mathcal{E}, z)$ be the number of $n \in \mathcal{E}$ all of whose prime factors have norm at least z, and let $S(\mathcal{E}, q)$ be the number of $n \in \mathcal{E}$ all of whose prime factors ϖ satisfy $\varpi \succcurlyeq q$. These are the sifting functions of Harman and Lewis [7, p. 121], and z and q are their *levels*. The function $S(\mathcal{E}_u, z)$ also sifts the prime factors of u. For $n \in \mathcal{E}_u$, the quotient n/u is the *cofactor* of n. We also sift the cofactor alone:

$$S'(\mathcal{E}_u, z) = \#\{n \in \mathcal{E}_u : |\varpi|^2 \geqslant z \text{ for all } \varpi \mid n/u\}, \tag{4}$$

and $S'(\mathcal{E}_u, q)$ is defined in the same way with the condition $\varpi \succcurlyeq q$. Thus $S(\mathcal{E}_u, z) = S'(\mathcal{E}_u, z)$ if every prime factor of u has norm at least z, and $S(\mathcal{E}_u, z) = 0$ otherwise. Classifying n/u by its least prime factor for $\prec$ gives Buchstab's identities

$$S'(\mathcal{E}_u, z) = S'(\mathcal{E}_u, w) - \sum_{w \leqslant |\varpi|^2 < z} S'(\mathcal{E}_{u\varpi}, \varpi), \qquad S'(\mathcal{E}_u, q) = S'(\mathcal{E}_u, w) - \sum_{\varpi \prec q,\ |\varpi|^2 \geqslant w} S'(\mathcal{E}_{u\varpi}, \varpi), \tag{5}$$

for $1 \leqslant w \leqslant z$ and for primes q with $|q|^2 \geqslant w$. These are Harman and Lewis [7, (13)], and the same identities hold for S in place of S' when $u = 1$.

The density of $\mathcal{A}$ in $\mathcal{B}$ is $2/(\pi W)$. A sum $\Sigma(\mathcal{E})$, defined for $\mathcal{E} \in \{\mathcal{A}, \mathcal{B}\}$, *has an asymptotic formula* if, for every $A > 0$,

$$\Sigma(\mathcal{A}) = \frac{2}{\pi W}\,\Sigma(\mathcal{B}) + \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big). \tag{6}$$

A sum has an asymptotic formula *uniformly* in a family when (6) holds with the same implied constant for every member.

Throughout, $\varepsilon > 0$ denotes a fixed small margin. An inequality between exponents holds *with margin* ε if it still holds after ε is added to its smaller side. Implied constants depend only on A, on θ, on ε, on the

exponent C_0 of the divisor bounds and on the number of variables. They never depend on dyadic boxes, intervals, levels or coefficients, which is the convention of Harman and Lewis [7, p. 121]. The letter C denotes a positive constant with the same dependence, not necessarily the same at each occurrence.

For real $y \geqslant 1$, let $j(y) = -1$ if $y \geqslant W$, and otherwise let $j(y) \geqslant 0$ be the integer with $WX^{-(j(y)+1)\theta} \leqslant y < WX^{-j(y)\theta}$. Following Harman and Lewis [7, p. 128], put

$$z(y) = \Big(\frac{X^\tau}{y}\Big)^{1/(j(y)+2)}. \tag{7}$$

Thus $z(y) = X^\tau/y$ for $y \geqslant W$. For $y < W$ and $j = j(y)$,

$$z(y) > \Big(\frac{X^\tau}{WX^{-j\theta}}\Big)^{1/(j+2)} = X^\theta.$$

3 Outline of the proof

In this section, $1/9 < \theta < 1/8$. In the notation of Section 2, the number of Gaussian primes in $\mathcal{A}$ is $S(\mathcal{A}, X^{1/2})$. Buchstab's identity (5) writes it as a combination of sums of the two kinds of Section 1. A sum of the first kind has an asymptotic formula in the sense of (6). Lemma 16 proves the lower bound (2) in the form (28), where $\Phi_\varepsilon(\theta)$ tends to the total loss $\Phi(\theta)$ of (14) as $\varepsilon \to 0$. Theorem 1 follows once $\Phi(\theta) < 1$.

The asymptotic formulas rest on the linear and bilinear estimates, Lemmas 1 and 2. The arithmetic information quoted from Harman and Lewis [7] enters the argument only through these two lemmas. Consider a sum over primes $\varpi_1, \dots, \varpi_K$ and a cofactor t with $\varpi_1 \cdots \varpi_K t \in \mathcal{E}$, where $|\varpi_i|^2$ is about X^{x_i}. For a set $\mathcal{S}$ of indices, put $x_\mathcal{S} = \sum_{i\in\mathcal{S}} x_i$. The linear estimate applies when t is free, that is, subject to no sieve condition. Through the fundamental lemma, Lemma 3, it also applies when t has no prime factor of norm below μ^4. In exponent form, it needs $\sum_i x_i \leqslant \tau$, or, for a partition of the indices into sets $\mathcal{U}$ and $\mathcal{V}$,

$$x_\mathcal{U} \leqslant \zeta, \qquad x_\mathcal{V} \leqslant \rho, \qquad \sum_i x_i \leqslant \nu.$$

The bilinear estimate applies at every level of sifting of t. It needs an index i_0 and a non-empty set $\mathcal{S}$ of other indices with

$$\sigma \leqslant x_\mathcal{S} \leqslant \tau - x_{i_0}.$$

Then the product of the primes of $\mathcal{S}$ has norm at least W, up to a constant factor. So does the quotient of $\varpi_1 \cdots \varpi_K t$ by this product and by ϖ_{i_0}. These two numbers are the variables u and v of Lemma 2, and ϖ_{i_0} is its prime variable ϖ. Lemma 8 states these conditions precisely, with margin ε. Its condition (P4) comes from Lemmas 4 and 5. It covers a cofactor sifted at a level that depends on the product $\varpi_1 \cdots \varpi_K$. Lemma 7 reduces a sum with coupled variables to sums that satisfy these conditions. The order condition $q \prec p$ and a sieve condition that depends on a prime are examples of such couplings.

The decomposition begins with two applications of Buchstab's identity. Let Π_1 be the set of primes p with $X^\theta \leqslant |p|^2 < X^{1/2}$ and $z(|p|^2) \leqslant |p|^2$, and let Π_2 be the set of the other primes with $X^\theta \leqslant |p|^2 < X^{1/2}$. For $\mathcal{E} \in \{\mathcal{A}, \mathcal{B}\}$,

$$S(\mathcal{E}, X^{1/2}) = S(\mathcal{E}, X^\theta) - \sum_{p\in\Pi_1} S\big(\mathcal{E}_p, z(|p|^2)\big) - \sum_{p\in\Pi_2} S(\mathcal{E}_p, p) + \sum_{p,q}{}^{*} S(\mathcal{E}_{pq}, q), \tag{8}$$

where $\sum^*$ runs over the pairs with $p \in \Pi_1$, $q \prec p$ and $|q|^2 \geqslant z(|p|^2)$. Lemma 13 proves (8). It gives the terms other than $\sum^*$ asymptotic formulas by Corollary 6 and Lemmas 1, 4 and 5. The rest of the argument concerns $\sum^*$. We write $|p|^2 = X^\alpha$, $|q|^2 = X^\beta$, $|r|^2 = X^\gamma$ and $|s|^2 = X^\delta$. For $0 < \alpha < 1/2$ let $\beta_0(\alpha) = \log z(X^\alpha)/\log X$, so that

$$\beta_0(\alpha) = \begin{cases} \tau - \alpha & \text{for } \alpha \geqslant \sigma, \\ (\tau - \alpha)/(j+2) & \text{for } \alpha < \sigma, \text{ where } j = j(X^\alpha). \end{cases}$$

Section 7 discards a set of pairs of $\sum^*$ with cofactor 1, whose sum over $\mathcal{B}$ is $\mathcal{O}(X/(\log X)^2)$. The other pairs have exponents in

$$\mathsf{D} = \big\{(\alpha,\beta) : \theta \leqslant \alpha < 1/2,\ \beta_0(\alpha) \leqslant \alpha,\ \beta_0(\alpha) \leqslant \beta < \min(\alpha, (1-\alpha)/2)\big\}. \tag{9}$$

Following Harman and Lewis [7, §5], we split D into Ranges 1 to 6, listed in Table 1, and each range into parts. At $\theta = 0.1129$, the boundaries in α are $\sigma = 0.3871$, $1/3$, $\rho = 0.29032$ and $\sigma - \theta = 0.2742$. On each part, the sum has an asymptotic formula, or identity (18) of Lemma 11 or identity (19) of Lemma 12 applies, or the whole sum is discarded. Each identity applies Buchstab's identity (5) to $S(\mathcal{A}_{pq}, q)$, which introduces a prime $r \prec q$, and then to $S(\mathcal{A}_{pqr}, r)$, which introduces a prime $s \prec r$. When an identity applies, its first and second sums have asymptotic formulas, and its third sum, of non-negative terms with the sign $+$, is discarded. The third sums are sums of $S(\mathcal{A}_{pqrs}, s)$ over primes $s \prec r \prec q \prec p$, so each of them is discarded over a four-dimensional region of exponents $(\alpha, \beta, \gamma, \delta)$. Section 7 defines the discarded regions D_k and E_k of Table 1. It checks the hypotheses of the lemmas on each part, with the inequalities of Appendix D.

Range	Exponents	Treatment of $\sum^*$	Discarded	Third sums
1	$\sigma \leqslant \alpha < 1/2$	asymptotic formula (Lemma 9)	D_1	—
2	$1/3 \leqslant \alpha < \sigma$	identity (18), case (a), and case (b) on R (Lemma 11)	D_2	E_3, E_R
3	$\rho \leqslant \alpha < 1/3$	identity (18), case (a) (Lemma 11)	D_4	E_5
4	$\sigma - \theta \leqslant \alpha < \rho$	identity (18), case (a) (Lemma 11)	D_6	E_7
5	$\alpha < \sigma - \theta \leqslant \alpha + \beta$	where $\alpha + \beta \geqslant \sigma$: $S_<$, and identity (18), case (a) (Lemma 11); where $\alpha + \beta < \sigma$: $S_<$, and identity (19) with $j = 0$ (Lemma 12)	D_8	E_9, E_{10}
6	$\alpha + \beta < \sigma - \theta$	identity (19) with $j = 1$ (Lemma 12)	—	E_{11}

Table 1: The ranges of the domain D of (9). For each range: the exponents of the pairs (p, q) of $\sum^*$; the treatment, where $S_<$ is the part of Lemma 11 or 12 with an asymptotic formula; the two-dimensional region where the whole sum is discarded; and the four-dimensional regions where the third sum of the identity is discarded. The regions are defined in Section 7. The index k of a region D_k or E_k is that of the loss φ_k of Harman and Lewis, and E_R has no counterpart there.

Three groupings give asymptotic formulas to parts of the discarded sums. The grouping in Range 2 acts on the region

$$\mathsf{R} = \big\{1/3 \leqslant \alpha < \sigma,\ \max(\zeta - \alpha, (\tau - \alpha)/2) \leqslant \beta \leqslant \rho/2\big\}. \tag{10}$$

Case (a) of Lemma 11 applies the linear estimate to the factorization $(pq) \cdot r$, which needs $\alpha + \beta \leqslant \zeta$. On R, where $\alpha + \beta \geqslant \zeta$, case (b) applies it to the factorization $p \cdot (qr)$. So on R, identity (18) replaces the discard of the whole sum, and only its third sum is discarded, over the region E_R.

Harman and Lewis discard the third sums of (18) and (19), except on a few subregions that they remove by further inequalities. Appendix E discusses these inequalities, the *cuts*. The other two groupings give parts of the third sums asymptotic formulas. An index set is written by its primes, as in $x_{\{p,q\}} = \alpha + \beta$. Define the following regions of $(\alpha, \beta, \gamma, \delta)$:

$$\begin{aligned}
\mathsf{G} &= \bigcup_{\emptyset \neq \mathcal{S} \subset \{p,q,r\}} \big\{\sigma \leqslant x_\mathcal{S} \leqslant \tau - \delta\big\},\\
\mathsf{I} &= \big\{\alpha + \beta + \gamma + \delta \leqslant \tau\big\} \cup \bigcup_{\mathcal{U} \sqcup \mathcal{V} = \{p,q,r,s\}} \big\{x_\mathcal{U} \leqslant \zeta,\ x_\mathcal{V} \leqslant \rho,\ \alpha + \beta + \gamma + \delta \leqslant \nu\big\},\\
\mathsf{L} &= \mathsf{I} \cap \bigcup_{\emptyset \neq \mathcal{S} \subset \{p,q,r,s\}} \big\{\sigma \leqslant x_\mathcal{S} \leqslant \tau - \delta\big\}.
\end{aligned} \tag{11}$$

The region G comes from the bilinear estimate applied to products of the five factors p, q, r, s, t, with s as the prime variable. The region L comes from a further Buchstab step in $S(\mathcal{A}_{pqrs}, s)$, at level μ^4, which introduces a prime $s' \prec s$. The step gives the difference (25) of a sum sifted at level μ^4 and a sum over s'. The sum over s' enters with the sign $-$, so both sums need asymptotic formulas. The linear estimate for $pqrs$ treats

the sum sifted at level μ^4 on I. The bilinear estimate with the prime s' treats the sum over s' on the union over $\mathcal{S}$ in the definition of L. Lemma 14 proves the formulas on G and L. The region L contains the polytope

$$\mathsf{P} = \big\{\alpha+\beta \leqslant \sigma,\ \gamma+\delta \leqslant \rho,\ \alpha+\beta+\delta \geqslant \sigma,\ \alpha+\beta+2\delta \leqslant \tau\big\}. \tag{12}$$

On P, the partition $\mathcal{U} = \{p,q\}$, $\mathcal{V} = \{r,s\}$ satisfies the conditions of I, since $\sigma < \zeta$ and $\sigma+\rho = \nu$. The set $\mathcal{S} = \{p,q,s\}$ satisfies $\sigma \leqslant x_{\mathcal{S}} \leqslant \tau-\delta$. Section 10 also evaluates the losses with P in place of L. Every region is then an explicit polytope, and the treatment of each four-dimensional box can be checked by hand.

For a two-dimensional region F and a four-dimensional region F', put

$$\Phi(\mathsf{F}) = \iint_{\mathsf{F}} \omega\Big(\frac{1-\alpha-\beta}{\beta}\Big)\frac{d\beta\,d\alpha}{\alpha\beta^2}, \qquad \Phi(\mathsf{F}') = \iiiint_{\mathsf{F}'} \omega\Big(\frac{1-\alpha-\beta-\gamma-\delta}{\delta}\Big)\frac{d\delta\,d\gamma\,d\beta\,d\alpha}{\alpha\beta\gamma\delta^2}, \tag{13}$$

where ω is Buchstab's function, defined by $\omega(u) = 0$ for $u < 1$ and

$$u\omega(u) = 1 + \int_1^{u-1} \omega(t)\,dt \qquad (u \geqslant 1).$$

Let $\Phi(\theta)$ be the total loss

$$\Phi(\theta) = \sum_{k\in\{1,2,4,6,8\}} \Phi(\mathsf{D}_k) + \sum_{k\in\{3,\mathsf{R},5,7,9,10,11\}} \Phi\big(\mathsf{E}_k \setminus (\mathsf{G}\cup\mathsf{L})\big), \tag{14}$$

with the regions of Section 7 and (11). Let $\Phi_{\mathsf{P}}(\theta)$ be defined in the same way with P in place of L.

Lemma 16 splits $\sum^*$ and the third sums into dyadic boxes. It applies Lemmas 9, 11, 12 and 14 to the boxes that lie inside a treated region with margin ε, and it bounds the other boxes by Lemma 15. Table 2 gives $\Phi(0.1129) = 0.9990033$, and Section 10 completes the proof.

4 Asymptotic formulas from the linear and bilinear estimates

Lemmas 1 and 2 below are the linear and bilinear estimates of Section 3, in the form used in this paper. Appendix A derives them from the arithmetic information of Harman and Lewis [7]. The other results of this section sift the cofactor. Lemma 3 sifts it at levels up to μ^4. Lemmas 4 and 5 sift the cofactors of the elements of $\mathcal{E}_n$ at a level $l(n)$ that depends on n. With Corollary 6, they extend Lemmas 8, 9 and 10 of Harman and Lewis [7] (Appendix E). Appendix B contains the proofs of Lemmas 3, 4 and 5.

Lemma 1 *Let $c_v \geqslant 0$ be divisor-bounded, and suppose that one of the following holds:*

(a) c_v is supported on $|v|^2 \leqslant X^\tau\mu^{-1}$;

(b) $c_v = \sum_{v=uu'} a_u b_{u'}$, where $a_u, b_{u'} \geqslant 0$ are divisor-bounded and supported on $u \sim U$, $u' \sim U'$, with $U \leqslant X^\zeta\mu^{-1}$, $U' \leqslant X^\rho\mu^{-1}$ and $UU' \leqslant X^\nu\mu^{-1}$.

Then $\sum_v c_v|\mathcal{E}_v|$ has an asymptotic formula.

The bilinear estimate of Harman and Lewis [7, Lemma 5 and (30)] needs a prime variable. Its proof applies when the prime is restricted to any interval of norms above $\mu^4/2$ (Remark 25).

Lemma 2 *Let $C_1 \geqslant 1$, let $a_u, b_v \geqslant 0$ be divisor-bounded and supported on $u \sim U$ and $v \sim V$ with $U, V \geqslant W/C_1$, and let $\mu^4/2 \leqslant Z_1 < Z_2$. Then*

$$\sum_{u,v} a_u b_v\, \#\{\varpi : Z_1 \leqslant |\varpi|^2 < Z_2,\ uv\varpi \in \mathcal{E}\} \tag{15}$$

has an asymptotic formula, uniformly in U, V, Z_1 and Z_2.

Lemma 3 below is a fundamental lemma over $\mathbb{Z}[i]$. Every level at which it is applied is at most $\mu^4 = X^{o(1)}$, so Brun's pure sieve suffices (Appendix B). For $z \geqslant 1$, let $\mathfrak{P}(z)$ be the product of the primes of norm less than z, and for a squarefree $\mathfrak{d}$ let $\mathfrak{o}(\mathfrak{d})$ be its number of prime factors.

Lemma 3 *Let $0 < \varepsilon_0 < 1$ and $M+1 = \lfloor (\varepsilon_0/4)(\log X)^{1/5} \rfloor$. Let $c_u \geqslant 0$ be divisor-bounded and supported on $|u|^2 \leqslant X$, and let z_0 be a function on the support of c with $1 \leqslant z_0(u) \leqslant \mu^4$. For squarefree $\mathfrak{d}$ put*

$$\mathfrak{b}_{\mathfrak{d}}^{\pm}(u) = 1\big[\mathfrak{d} \mid \mathfrak{P}(z_0(u)),\ \mathfrak{o}(\mathfrak{d}) \leqslant M,\ (-1)^{\mathfrak{o}(\mathfrak{d})} = \pm 1\big], \qquad \mathfrak{b}_{\mathfrak{d}}^{0}(u) = 1\big[\mathfrak{d} \mid \mathfrak{P}(z_0(u)),\ \mathfrak{o}(\mathfrak{d}) = M+1\big],$$

and put $\mathfrak{b}_{\mathfrak{d}}^{+}(u) = \mathfrak{b}_{\mathfrak{d}}^{-}(u) = \mathfrak{b}_{\mathfrak{d}}^{0}(u) = 0$ for $\mathfrak{d}$ not squarefree. Suppose that for each $\mathfrak{b} \in \{\mathfrak{b}^+, \mathfrak{b}^-, \mathfrak{b}^0\}$ the sum $\sum_{u,\mathfrak{d}} c_u \mathfrak{b}_{\mathfrak{d}}(u) |\mathcal{E}_{u\mathfrak{d}}|$ has an asymptotic formula. Then $\sum_u c_u S'(\mathcal{E}_u, z_0(u))$ has an asymptotic formula. The hypothesis holds by Lemma 1(a) when c_u is supported on $|u|^2 \leqslant X^{\tau-\varepsilon_0}\mu^{-1}$. It holds by Lemma 1(b) when z_0 is constant and c has the form $\sum_{u=u_1u_2} a_{u_1} b_{u_2}$ of Lemma 1(b), with the bounds on U' and UU' lowered by the factor X^{ε_0}.

Lemmas 4 and 5 are stated for S' and for $a_n \geqslant 0$. This loses nothing. For a divisor-bounded $a_n \geqslant 0$ and a function l,

$$\sum_n a_n S(\mathcal{E}_n, l(n)) = \sum_n a'_n S'(\mathcal{E}_n, l(n)),$$

where $a'_n = a_n$ if every prime factor of n has norm at least $l(n)$, and $a'_n = 0$ otherwise. A divisor-bounded complex sequence is a combination of 4 non-negative divisor-bounded sequences.

Lemma 4 *Let $C_l \geqslant 1$. Let $a_n \geqslant 0$ be divisor-bounded and supported on $W/C_l \leqslant |n|^2 \leqslant X^{\tau-\varepsilon}$, and let l be any function on the support of a with $1 \leqslant l(n) \leqslant C_l X^{\tau}/|n|^2$. Then $\sum_n a_n S'(\mathcal{E}_n, l(n))$ has an asymptotic formula, uniformly in a and l.*

Lemma 5 *Let $C_l \geqslant 1$. Let $a_n \geqslant 0$ be divisor-bounded and supported on $1 \leqslant |n|^2 < W$, and let l be any function on the support of a with $1 \leqslant l(n) \leqslant C_l\, z(|n|^2)$. Then $\sum_n a_n S'(\mathcal{E}_n, l(n))$ has an asymptotic formula, uniformly in a and l.*

Corollary 6 *The sum $S(\mathcal{E}, X^{\theta})$ has an asymptotic formula.*

Proof. The sum $S(\mathcal{E}, X^{\theta}) = S'(\mathcal{E}_1, X^{\theta})$ is the case $a_n = 1[n=1]$, $l \equiv X^{\theta}$ of Lemma 5. Its hypothesis holds, since $z(1) > X^{\theta}$ by the bound after (7). □

5 Removing the conditions that couple the variables

The sums of Harman's sieve run over products of primes with order conditions between them, conditions on products of their norms, and sieve conditions on the cofactor that depend on one of the primes. The bilinear estimate of Lemma 2 admits none of these conditions between its variables. This section removes them with short intervals. Every condition is constant on most Cartesian products of short intervals, and the sums over the other products are small. Lemma 7 reduces a sum with such conditions to sums over single Cartesian products of short intervals, and Lemma 8 gives these sums asymptotic formulas under the conditions of Section 3.

Let $H = (\log X)^B$ with B fixed, as in Lemma 17 of Appendix A, and put $\xi_\iota = (1+1/H)^\iota$ for integers $\iota \geqslant 0$. The *short intervals* are $J_\iota = [\xi_\iota, \xi_{\iota+1})$. For a prime q, let ι_q be the index with $|q|^2 \in J_{\iota_q}$.

A *configuration* $\mathcal{C}$ consists of the following data:

(i) prime variables $\varpi_1, \dots, \varpi_k$, with k fixed and $|\varpi_i|^2 \in \mathcal{I}_i$ for intervals $\mathcal{I}_i \subset [\mu^4, X]$, the *ranges* of the variables. A *tuple* is a choice $\varpi = (\varpi_1, \dots, \varpi_k)$ of these primes;

(ii) a cofactor t, which is a Gaussian integer;

(iii) finitely many *monomial constraints* c, each of the form $m_c(\varpi) < Z_c$ or $m_c(\varpi) \leqslant Z_c$, where $m_c(\varpi) = \prod_i |\varpi_i|^{2e_{c,i}}$ with integers $|e_{c,i}| \leqslant 2$, not all 0, and $Z_c > 0$. We write $1_c(\varpi)$ for the indicator that ϖ satisfies c. A lower bound $m_c(\varpi) \geqslant Z_c$ is the constraint $m_c(\varpi)^{-1} \leqslant Z_c^{-1}$. An order constraint $\varpi_i \prec \varpi_{i'}$ is the monomial constraint with $m_c(\varpi) = |\varpi_i|^2/|\varpi_{i'}|^2$ and $Z_c = 1$, in which the case $m_c(\varpi) = 1$ is decided by $\prec$;

(iv) a *sieve condition* on t, with indicator $\Xi(\varpi, t)$, of type (F), (C), (S) or (M): (F) none; (C) every prime factor of t has norm at least μ^4; (S) every prime factor ϖ' of t satisfies $\varpi' \succcurlyeq \varpi_{i_s}$, for a fixed index i_s; (M) every prime factor of t has norm at least $m_s(\varpi) = Z_s \prod_i |\varpi_i|^{2f_i}$, with fixed $Z_s > 0$ and fixed reals $|f_i| \leqslant 2$.

On every tuple that satisfies all the constraints, we require $\varpi_{i_s} \preccurlyeq \varpi_i$ for all i in type (S), and $\mu^4 \leqslant m_s(\varpi) \leqslant \min_i |\varpi_i|^2$ in type (M). The subscript s of i_s, m_s and Z_s refers to the sieve condition, not to a prime s. The sum of $\mathcal{C}$ is

$$\Sigma_{\mathcal{C}}(\mathcal{E}) = \sum_{\varpi_1,\dots,\varpi_k} \sum_t 1[\varpi_1 \cdots \varpi_k t \in \mathcal{E}] \prod_i 1[|\varpi_i|^2 \in \mathcal{I}_i] \prod_c 1_c(\varpi)\, \Xi(\varpi, t).$$

A configuration has the type of its sieve condition, and we identify it with its sum $\Sigma_{\mathcal{C}}$. For example, for fixed primes p, q, r, s, the sum $S'(\mathcal{E}_{pqrs}, s)$ counts the cofactors t with $pqrst \in \mathcal{E}$ whose prime factors are all $\succcurlyeq s$. A sum of $S'(\mathcal{E}_{pqrs}, s)$ over primes in given ranges with $s \prec r \prec q$ is therefore a configuration with $k = 4$, the order constraints $s \prec r$ and $r \prec q$, and the sieve condition of type (S) with i_s the index of s.

For an index vector $\iota = (\iota_1, \dots, \iota_k)$, let $\mathcal{T}(\iota)$ be the set of tuples with $|\varpi_i|^2 \in \mathcal{I}_i \cap J_{\iota_i}$ for all i. For a constant $y \geqslant 1$ and an integer κ put

$$\mathcal{P}_\iota(\mathcal{E}; y) = \sum_{\varpi \in \mathcal{T}(\iota)} S'(\mathcal{E}_{\varpi_1 \cdots \varpi_k}, y), \qquad \mathcal{P}'_{\iota,\kappa}(\mathcal{E}) = \sum_{\varpi \in \mathcal{T}(\iota)} \sum_{|\varpi'|^2 \in J_\kappa} |\mathcal{E}_{\varpi_1 \cdots \varpi_k \varpi'}|,$$

where ϖ' runs over primes. For $y = 1$ the cofactor is free, since $S'(\mathcal{E}_u, 1) = |\mathcal{E}_u|$. These sums are the *pieces* of $\mathcal{C}$.

The index vector ι *satisfies* a constraint c if every tuple of $\mathcal{T}(\iota)$ satisfies c, and *straddles* c if $\mathcal{T}(\iota)$ contains tuples that satisfy c and tuples that do not. The *level* of ι is

$$y_\iota = \begin{cases} 1 & \text{in type (F)}, \\ \mu^4 & \text{in type (C)}, \\ \xi_{\iota_{i_s}} & \text{in type (S)}, \\ \inf_{\mathcal{T}(\iota)} m_s & \text{in type (M)}, \end{cases}$$

and its *window* is

$$\mathcal{W}(\iota) = \begin{cases} \emptyset & \text{in types (F) and (C)}, \\ \{\iota_{i_s}\} & \text{in type (S)}, \\ \{\kappa : J_\kappa \cap [\inf_{\mathcal{T}(\iota)} m_s, \sup_{\mathcal{T}(\iota)} m_s) \neq \emptyset\} & \text{in type (M)}. \end{cases}$$

Since m_s varies by at most the factor $(1 + 1/H)^{2k}$ on $\mathcal{T}(\iota)$, the window has at most $2k + 2$ elements. If ι satisfies all constraints, the requirements on types (S) and (M) give $y_\iota \leqslant \min_i |\varpi_i|^2$ on $\mathcal{T}(\iota)$ and $\xi_\kappa \geqslant \mu^4/2$ for $\kappa \in \mathcal{W}(\iota)$.

Lemma 7 *Let $\mathcal{C}$ be a configuration, and suppose that the following sums have asymptotic formulas, uniformly in ι and κ:*

(a) $\mathcal{P}_\iota(\mathcal{E}; y_\iota)$ for every ι that satisfies all constraints;

(b) $\mathcal{P}_\iota(\mathcal{E}; 1)$ for every ι that straddles some constraint;

(c) $\mathcal{P}'_{\iota,\kappa}(\mathcal{E})$ for every ι that satisfies all constraints and every $\kappa \in \mathcal{W}(\iota)$.

Then $\Sigma_{\mathcal{C}}$ has an asymptotic formula.

The pieces in (b) of Lemma 7 are the *straddling pieces* of $\mathcal{C}$, and the pieces in (c) are its *window pieces*. Appendix B proves Lemma 7.

The hypotheses of Lemma 7 are checked with Lemma 8 below. The primes of a piece are $\varpi_1, \dots, \varpi_K$, where $K = k$ for $\mathcal{P}_\iota$, and $K = k + 1$ with $\varpi_{k+1} = \varpi'$ for $\mathcal{P}'_{\iota,\kappa}$. Their norms lie in the intervals $I_i = \mathcal{I}_i \cap J_{\iota_i}$ for $i \leqslant k$ and $I_{k+1} = J_\kappa$, which satisfy $I_i \subset [\mu^4/2, X]$ and $\sup I_i \leqslant (1 + 1/H) \inf I_i$. The cofactor t of a piece

is sifted at a constant level y, with $y = 1$ for $\mathcal{P}'_{\iota,\kappa}$. Let $x_i = \log(\inf I_i)/\log X$ be the exponents of the piece. A condition that holds at (x_i) with margin ε holds with margin $\varepsilon/2$ for every tuple of the piece, once X is large, since each $\log|\varpi_i|^2/\log X$ lies within $\mathcal{O}(1/(H\log X))$ of x_i. When the primes of a piece have names, an index set is written by its primes, as in $\mathcal{U} = \{p\}$, and i_0 by its prime, as in $\varpi_{i_0} = r$.

Lemma 8 *Let $C_1 \geqslant 1$ be fixed. A piece has an asymptotic formula, uniformly in ι and κ, if one of the following holds.*

(P1) $y = 1$, *and either* $\sum_i x_i \leqslant \tau - \varepsilon$, *or the indices split as* $\mathcal{U} \sqcup \mathcal{V}$ *with*

$$x_{\mathcal{U}} \leqslant \zeta - \varepsilon, \qquad x_{\mathcal{V}} \leqslant \rho - \varepsilon, \qquad \sum_i x_i \leqslant \nu - \varepsilon.$$

(P2) $y = \mu^4$, *and the exponents satisfy one of the alternatives of (P1).*

(P3) There are an index i_0 and a partition of the other indices into $\mathcal{S} \neq \emptyset$ and $\mathcal{S}'$ such that

$$\inf\Big|\prod_{i\in\mathcal{S}} \varpi_i\Big|^2 \geqslant \frac{W}{C_1}, \qquad \frac{X/2}{\sup|\varpi_{i_0}\prod_{i\in\mathcal{S}}\varpi_i|^2} \geqslant \frac{W}{C_1},$$

with the infimum and the supremum over the tuples of the piece.

(P4) $y \leqslant \min_i \inf I_i$, *and every product $n = \varpi_1 \cdots \varpi_K$ of the piece satisfies*

$$W/C_1 \leqslant |n|^2 \leqslant X^{\tau-\varepsilon}, \quad y \leqslant C_1 X^\tau/|n|^2, \qquad \text{or} \qquad |n|^2 < W, \quad y \leqslant C_1\, z(|n|^2).$$

Proof. In (P1), the piece is $\sum_v a_v|\mathcal{E}_v|$, where a_v is the number of tuples of the piece with product v. This a_v is non-negative, at most $d(v)^K$, and supported on $|v|^2 \leqslant X^{\sum x_i + o(1)}$. In the second case $a_v = \sum_{v=uu'} a'_u b_{u'}$, where a'_u is the number of subtuples $(\varpi_i)_{i\in\mathcal{U}}$ with product u, and $b_{u'}$ is the number of subtuples $(\varpi_i)_{i\in\mathcal{V}}$ with product u'. Each is supported on at most two dyadic ranges. Lemma 1 applies, since the margin $\varepsilon/2$ absorbs the factor μ^{-1}.

In (P2), the piece is $\sum_u a_u S'(\mathcal{E}_u, \mu^4)$, where a_u is the number of tuples of the piece with product u, and Lemma 3 applies with $\varepsilon_0 = \varepsilon/4$. In the second case the divisors $\mathfrak{d}$ are attached to the part indexed by $\mathcal{V}$, whose range is split into $\mathcal{O}(\log X)$ dyadic ranges.

In (P3), the piece is

$$\sum_{u,v} a_u b_v \,\#\big\{\varpi_{i_0} : |\varpi_{i_0}|^2 \in I_{i_0},\ uv\varpi_{i_0} \in \mathcal{E}\big\},$$

where a_u is the number of subtuples $(\varpi_i)_{i\in\mathcal{S}}$ with product u, and b_v is the number of pairs of a subtuple $(\varpi_i)_{i\in\mathcal{S}'}$ and a cofactor t sifted at level y with product v. Both are non-negative and divisor-bounded. The variable u lies in at most two dyadic ranges and v in at most four. Both variables are at least W/C_1. The norm of ϖ_{i_0} ranges over $I_{i_0} \subset [\mu^4/2, X]$, so Lemma 2 applies.

In (P4), the piece is $\sum_n a_n S'(\mathcal{E}_n, y)$, where a_n counts the tuples with product n. Lemma 4 applies to the n of the first alternative and Lemma 5 to the others, both with the constant level y. □

6 Sums over two primes

The lemmas of this section treat the sum $\sum^*$ of (8) on dyadic boxes. They supply the treatments of Table 1, and Section 7 checks their hypotheses on each range. In the proof of Lemma 11, Lemma 10 gives the second sum of (18) an asymptotic formula.

Throughout, $p \sim P$ and $q \sim Q$, where P and Q are powers of 2. The pairs (p, q) with $p \sim P$ and $q \sim Q$ form a *box*. Its *exponent rectangle* is the set of the pairs (α, β) with $X^\alpha \in [P, 2P)$ and $X^\beta \in [Q, 2Q)$, and the distance of a box from a set of exponents is that of its exponent rectangle.

By the definition (4) of S', $S(\mathcal{A}_u, y) = S'(\mathcal{A}_u, y)$ for every sifting function below whose level y does not exceed the norm of any prime factor of u. In the same way, $S(\mathcal{A}_{pq}, q) = S'(\mathcal{A}_{pq}, q)$ if $q \preccurlyeq p$.

Lemma 9 *Suppose that $W \leqslant P \leqslant X^{\zeta-\varepsilon}$, $Q \geqslant PX^{-2\theta}$ and $PQ \leqslant X^{\nu-\varepsilon}$. Then*

$$\sum_{p\sim P,\ q\sim Q} S(\mathcal{A}_{pq}, q)$$

has an asymptotic formula.

Proof. Since $Q \leqslant X^{\nu-\varepsilon}/W = X^{\rho-\varepsilon} < W \leqslant P$, the condition $q \prec p$ holds automatically. Buchstab's identity (5) at level μ^4 gives

$$\sum_{p,q} S'(\mathcal{A}_{pq}, q) = \sum_{p,q} S'(\mathcal{A}_{pq}, \mu^4) - \sum_{p,q} \sum_{r\prec q,\ |r|^2\geqslant\mu^4} S'(\mathcal{A}_{pqr}, r).$$

The sum of $S'(\mathcal{A}_{pq}, \mu^4)$ is a configuration of type (C) without constraints. Its pieces fall under (P2) of Lemma 8 with $\mathcal{U} = \{p\}$ and $\mathcal{V} = \{q\}$, since $P \leqslant X^{\zeta-\varepsilon}$, $Q \leqslant X^{\rho-\varepsilon}$ and $PQ \leqslant X^{\nu-\varepsilon}$.

In the sum of $S'(\mathcal{A}_{pqr}, r)$, a term with $|r|^4 \geqslant X/(PQ)$ vanishes. Indeed, a cofactor $t \neq 1$ has $|t|^2 \geqslant |r|^2$, so $|pqrt|^2 \geqslant PQ|r|^4 \geqslant X$. The cofactor $t = 1$ requires $|r|^2 \geqslant X/(8PQ)$, and $X/(8PQ) > 2Q$ because $PQ^2 \leqslant X^{\nu+\rho-2\varepsilon}$ and $\nu + \rho = (2 + 8\theta)/3 < 1$. So $|r|^2 > 2Q > |q|^2$, which contradicts $r \prec q$. Hence the sum of $S'(\mathcal{A}_{pqr}, r)$ is the configuration with variables p, q, r, the range $[\mu^4, (X/(PQ))^{1/2})$ for $|r|^2$, the constraint $r \prec q$, and the sieve condition of type (S) with i_s the index of r. Type (S) requires $r \preccurlyeq p$, which holds because $(X/(PQ))^{1/2} \leqslant X^{\tau}/P \leqslant X^{2\theta} < P$. All of its pieces fall under (P3) with $\mathcal{S} = \{p\}$, $\varpi_{i_0} = r$, and $\mathcal{S}' = \{q\}$, or $\mathcal{S}' = \{q, \varpi'\}$ for the window pieces. Indeed, $|p|^2 \geqslant W$, and the quotient $(X/2)/\sup|pr|^2$ of (P3) satisfies

$$\frac{X/2}{2P(X/(PQ))^{1/2}(1 + 1/H)} \geqslant \frac{(XQ/P)^{1/2}}{5} \geqslant \frac{W}{5},$$

since $Q \geqslant PX^{-2\theta}$. Lemma 7 completes the proof. □

Lemma 10 *Let $Q \leqslant 2P$, $PQ \geqslant W$ and $X^{\tau}/(PQ) \geqslant X^{\varepsilon}$, and let the norm of q range over an interval $\mathcal{Q} \subset [Q, 2Q)$ with $\inf \mathcal{Q} \geqslant X^{\tau}/(PQ)$. Assume one of*

$$\text{(a) } PQ \leqslant X^{\zeta-\varepsilon},\ Q \leqslant X^{\rho-\varepsilon},\ PQ^2 \leqslant X^{\nu-\varepsilon}; \qquad \text{(b) } P \leqslant X^{\zeta-\varepsilon},\ Q^2 \leqslant X^{\rho-\varepsilon},\ PQ^2 \leqslant X^{\nu-\varepsilon}. \tag{16}$$

Let $\mathcal{I} \subset [X^{\tau}/(PQ), 2Q)$ be an interval. Then

$$\sum_{p\sim P,\ |q|^2\in\mathcal{Q},\ q\prec p} \sum_{|r|^2\in\mathcal{I},\ r\prec q} S\big(\mathcal{A}_{pqr}, X^{\tau}/(PQ)\big) \tag{17}$$

has an asymptotic formula.

Proof. All of p, q, r have norm at least $X^{\tau}/(PQ) \geqslant X^{\varepsilon} \geqslant \mu^4$, so Buchstab's identity (5) gives

$$S\big(\mathcal{A}_{pqr}, X^{\tau}/(PQ)\big) = S'(\mathcal{A}_{pqr}, \mu^4) - \sum_{\mu^4\leqslant|s|^2<X^{\tau}/(PQ)} S'(\mathcal{A}_{pqrs}, s).$$

Summed as in (17), the term $S'(\mathcal{A}_{pqr}, \mu^4)$ is the configuration with variables p, q, r, ranges $[P, 2P)$, $\mathcal{Q}$, $\mathcal{I}$, constraints $q \prec p$ and $r \prec q$, and the sieve condition of type (C). The sum of $S'(\mathcal{A}_{pqrs}, s)$ is the configuration with the further variable s, of range $[\mu^4, X^{\tau}/(PQ))$, the same constraints, and the sieve condition of type (S) with i_s the index of s. Every tuple that satisfies the constraints has $s \prec r \prec q \prec p$, as type (S) requires. We apply Lemma 7 to both.

For the configuration of type (C), the pieces of hypothesis (a) of Lemma 7 fall under (P2) with $\mathcal{U} = \{p, q\}$ and $\mathcal{V} = \{r\}$ in case (a) of (16), and with $\mathcal{U} = \{p\}$ and $\mathcal{V} = \{q, r\}$ in case (b). In cases (a) and (b) respectively, the part indexed by $\mathcal{U}$ has norm at most $4PQ$ or $2P$, and the part indexed by $\mathcal{V}$ at most $2Q$ or $4Q^2$. The product has norm at most $8PQ^2$, so (16) gives the conditions of (P1) with margin $\varepsilon/2$. The straddling pieces fall under (P1) with the same split, and there are no window pieces.

For the configuration of type (S), all pieces fall under (P3) with $\mathcal{S} = \{p, q\}$, $\varpi_{i_0} = s$, and $\mathcal{S}' = \{r\}$, or $\mathcal{S}' = \{r, \varpi'\}$ for the window pieces. Indeed, $|pq|^2 \geqslant PQ \geqslant W$. On every piece, $|s|^2 < (1 + 1/H)X^{\tau}/(PQ)$, so $(X/2)/\sup|pqs|^2 \geqslant W/9$. □

Lemma 11 *Suppose that $X^{\varepsilon} \leqslant Q \leqslant 2P$ and $W \leqslant PQ \leqslant X^{\tau-\varepsilon}$. Split*

$$\sum_{p\sim P,\, q\sim Q,\, q\prec p} S(\mathcal{A}_{pq}, q) = S_{<} + S_{\geqslant}$$

according as $|q|^2 < X^{\tau}/(PQ)$ or $|q|^2 \geqslant X^{\tau}/(PQ)$. Then $S_{<}$ has an asymptotic formula, and

$$S_{\geqslant} = \sum_{p,q} S\big(\mathcal{A}_{pq}, X^{\tau}/(PQ)\big) - \sum_{p,q} \sum_{\substack{r\prec q \\ |r|^2\geqslant X^{\tau}/(PQ)}} S\big(\mathcal{A}_{pqr}, X^{\tau}/(PQ)\big) + \sum_{p,q} \sum_{\substack{s\prec r\prec q \\ |s|^2\geqslant X^{\tau}/(PQ)}} S(\mathcal{A}_{pqrs}, s), \quad (18)$$

with the outer sums over $p\sim P$, $q\sim Q$, $q\prec p$, $|q|^2 \geqslant X^{\tau}/(PQ)$. If moreover (a) or (b) of (16) holds, then the first and second sums on the right of (18) have asymptotic formulas.

Proof. The sum $S_{<}$ is the configuration with variables p, q, ranges $[P, 2P)$ and $[Q, \min(2Q, X^{\tau}/(PQ)))$, the constraint $q \prec p$, and the sieve condition of type (S) with i_s the index of q. In Lemma 7, its pieces of hypothesis (a) fall under (P4), with $\varepsilon/2$ in place of ε, for $n = pq$ and the level $\xi_{\iota_q} \leqslant |q|^2$. Indeed, $W \leqslant |n|^2 \leqslant 4PQ \leqslant X^{\tau-\varepsilon/2}$ and $\xi_{\iota_q} < X^{\tau}/(PQ) \leqslant 4X^{\tau}/|n|^2$. Its straddling pieces, with p and q in the same short interval, fall under (P1), since $|pq|^2 \leqslant 4PQ$. Its window pieces, with ϖ' in the short interval of q, fall under (P3) with $\mathcal{S} = \{p, q\}$, $\varpi_{i_0} = \varpi'$ and $\mathcal{S}' = \emptyset$. Indeed, $|pq|^2 \geqslant W$, and

$$\frac{X/2}{4PQ\cdot(1+1/H)\,X^{\tau}/(PQ)} \geqslant \frac{W}{9}.$$

For $|q|^2 \geqslant X^{\tau}/(PQ)$, Buchstab's identity (5), applied first to $S(\mathcal{A}_{pq}, q)$ and then to each $S(\mathcal{A}_{pqr}, r)$ with $|r|^2 \geqslant X^{\tau}/(PQ)$, gives (18). The first sum on its right is $\sum_n a_n S(\mathcal{A}_n, X^{\tau}/(PQ))$ with $n = pq$ and a_n the number of pairs (p, q) in the range of the outer sums with $pq = n$. The level $X^{\tau}/(PQ)$ does not exceed $|q|^2 \leqslant |p|^2$, and Lemma 4 applies with $\varepsilon/2$ in place of ε, by the bounds shown for $S_{<}$. The second sum is the case $\mathcal{Q} = [\max(Q, X^{\tau}/(PQ)), 2Q)$ and $\mathcal{I} = [X^{\tau}/(PQ), 2Q)$ of Lemma 10, whose hypothesis $X^{\tau}/(PQ) \geqslant X^{\varepsilon}$ holds because $PQ \leqslant X^{\tau-\varepsilon}$. □

Lemma 12 *Let $j \in \{0, 1\}$ and $0 < 4\varepsilon < 5\theta/2 - 1/4$. Suppose that $X^{\varepsilon} \leqslant Q \leqslant 2P$ and $WX^{-(j+1)\theta} \leqslant PQ < WX^{-j\theta}$, so that $j(PQ) = j$ and $z(PQ) = (X^{\tau}/(PQ))^{1/(j+2)}$. Split*

$$\sum_{p\sim P,\, q\sim Q,\, q\prec p} S(\mathcal{A}_{pq}, q) = S_{<} + S_{\geqslant}$$

according as $|q|^2 < z(PQ)$ or $|q|^2 \geqslant z(PQ)$. Then $S_{<}$ has an asymptotic formula, and

$$\begin{aligned} S_{\geqslant} = {} & \sum_{p,q} S\big(\mathcal{A}_{pq}, z(PQ)\big) - \sum_{p,q} \sum_{\substack{r\prec q \\ |r|^2\geqslant z(PQ)}} S\Big(\mathcal{A}_{pqr}, \Big(\frac{X^{\tau}}{PQ|r|^2}\Big)^{1/(j+1)}\Big) \\ & + \sum_{p,q} \sum_{\substack{r\prec q \\ |r|^2\geqslant z(PQ)}} \sum_{\substack{s\prec r \\ |s|^{2(j+1)}\geqslant X^{\tau}/(PQ|r|^2)}} S(\mathcal{A}_{pqrs}, s), \end{aligned} \quad (19)$$

with the outer sums over $p\sim P$, $q\sim Q$, $q\prec p$, $|q|^2 \geqslant z(PQ)$. The first and second sums on the right of (19) have asymptotic formulas.

Proof. For $|q|^2 \geqslant z(PQ)$, Buchstab's identity (5) gives

$$S(\mathcal{A}_{pq}, q) = S\big(\mathcal{A}_{pq}, z(PQ)\big) - \sum_{\substack{r\prec q \\ |r|^2\geqslant z(PQ)}} S(\mathcal{A}_{pqr}, r).$$

For such r, the sifting level in the second sum of (19) is at most $|r|^2$, because $|r|^{2(j+2)} \geqslant z(PQ)^{j+2} = X^{\tau}/(PQ)$. So (5) applies again to $S(\mathcal{A}_{pqr}, r)$, and (19) follows.

Since $Q^2 \leqslant 2PQ$, every product of p, q and possibly a further prime of norm at most $4Q$ has norm at most $32(PQ)^{3/2} < 32X^{3\sigma/2}$. Since $3\sigma/2 = \tau - (5\theta/2 - 1/4)$ and $4\varepsilon < 5\theta/2 - 1/4$, it follows that, for every prime q' with $|q'|^2 \leqslant 4Q$,

$$|pq|^2|q'|^2 \leqslant X^{\tau-3\varepsilon}. \tag{20}$$

If $|r|^2 \geqslant z(PQ)$, then $|pqr|^2 \geqslant PQ\, z(PQ) = (PQ)^{(j+1)/(j+2)}X^{\tau/(j+2)}$, and the lower bound for PQ gives

$$|pqr|^2 \geqslant WX^{\theta/2} \quad (j=0), \qquad |pqr|^2 \geqslant WX^{-2\theta/3} \quad (j=1). \tag{21}$$

For $|r|^2 < 2Q$, the sifting level in the second sum of (19) satisfies

$$\Big(\frac{X^\tau}{PQ|r|^2}\Big)^{1/(j+1)} \geqslant \Big(\frac{X^\tau}{8PQ^2}\Big)^{1/(j+1)} \geqslant \frac{X^{(\tau-3\sigma/2)/2}}{8} \geqslant X^{2\varepsilon} \geqslant \mu^4. \tag{22}$$

The sum $S_<$ is the configuration with variables p, q, ranges $[P, 2P)$ and $[Q, \min(2Q, z(PQ)))$, the constraint $q \prec p$, and the sieve condition of type (S) with i_s the index of q. In Lemma 7, a piece of hypothesis (a) falls under (P4) with $n = pq$ and $y = \xi_{\iota_q} \leqslant |q|^2 < z(PQ)$. If $|n|^2 < W$, then $j(|n|^2) \leqslant j$ because $|n|^2 \geqslant PQ \geqslant WX^{-(j+1)\theta}$, so

$$z(|n|^2) \geqslant \Big(\frac{X^\tau}{4PQ}\Big)^{1/(j+2)} > \frac{z(PQ)}{4} > \frac{y}{4},$$

which is the second alternative of (P4). If $|n|^2 \geqslant W$, which is possible only for $j = 0$, then $|n|^2 \leqslant 4W \leqslant X^{\tau-\varepsilon}$ and

$$y < z(PQ) = \Big(\frac{X^\tau}{PQ}\Big)^{1/2} \leqslant \frac{X^\tau}{4PQ} \leqslant \frac{X^\tau}{|n|^2},$$

which is the first alternative. The straddling pieces and the window pieces fall under (P1) by (20).

The first sum of (19) is $\sum_n a_n S(\mathcal{A}_n, z(PQ))$ with $n = pq$ and a_n the number of pairs (p, q) in the range of the outer sums with $pq = n$. The level $z(PQ)$ does not exceed $|q|^2 \leqslant |p|^2$, and Lemma 5 (for $|n|^2 < W$) or Lemma 4 (for $|n|^2 \geqslant W$) applies with this constant level, exactly as for $S_<$.

The second sum of (19) is the configuration with variables p, q, r, ranges $[P, 2P)$, $[\max(Q, z(PQ)), 2Q)$ and $[z(PQ), 2Q)$, constraints $q \prec p$ and $r \prec q$, and the sieve condition of type (M) with

$$m_s(p,q,r) = \Big(\frac{X^\tau}{PQ}\Big)^{1/(j+1)}|r|^{-2/(j+1)}.$$

By (22) and the bound $m_s(p,q,r) \leqslant |r|^2$ shown above, the requirement of type (M) holds. In Lemma 7, a piece of hypothesis (a) has the level $y = \inf m_s$ over the piece, and falls under (P4) with $n = pqr$. For $j = 0$, every n has $|n|^2 \geqslant W$ by (21), $|n|^2 \leqslant X^{\tau-3\varepsilon}$ by (20), and $y \leqslant 4X^\tau/|n|^2$ because $|n|^2 < 4PQ\,\xi_{\iota_r+1}$. This is the first alternative of (P4). For $j = 1$, an n with $|n|^2 < W$ has $|n|^2 \geqslant WX^{-2\theta/3}$ by (21). Hence $j(|n|^2) = 0$ and $z(|n|^2) = (X^\tau/|n|^2)^{1/2} > y/2$, which is the second alternative of (P4). An n with $|n|^2 \geqslant W$ satisfies the first alternative, since $y \leqslant y^2 < 4X^\tau/|n|^2$.

The straddling pieces fall under (P1) by (20). A window piece adds a prime ϖ' with $|\varpi'|^2$ within the factor $(1 + 1/H)^2$ of $m_s(p,q,r)$. For $j = 1$ it falls under (P1), because $|pqr\varpi'|^2 \leqslant 8(X^\tau|pqr|^2)^{1/2} \leqslant X^{\tau-\varepsilon}$ by (20). For $j = 0$ it falls under (P3) with $\mathcal{S} = \{p, q, r\}$, $\varpi_{i_0} = \varpi'$ and $\mathcal{S}' = \emptyset$. Indeed, $|pqr|^2 \geqslant W$ by (21), and $|pqr\varpi'|^2 \leqslant 5X^\tau$ on the piece, so $(X/2)/\sup|pqr\varpi'|^2 \geqslant W/10$. □

7 The decomposition

This section proves identity (8), defines the regions of Table 1, and checks the treatment of each range. From now on, $1/9 < \theta < 1/8$.

Lemma 13 *Let Π_1, Π_2 and $\sum^*$ be as in Section 3. Identity (8) holds for $\mathcal{E} \in \{\mathcal{A}, \mathcal{B}\}$. The terms on the right of (8) other than $\sum^*$ have asymptotic formulas. A prime p with $|p|^2 = X^\alpha$ lies in Π_2 exactly when $\alpha \in [\max(\theta, \sigma - 3\theta), \tau/5)$ or $\alpha \in [\sigma - 2\theta, \tau/4)$. The interval $[\sigma - 2\theta, \tau/4)$ is empty for $\theta \leqslant 3/26$.*

Proof. Buchstab's identity (5) gives

$$S(\mathcal{E}, X^{1/2}) = S(\mathcal{E}, X^{\theta}) - \sum_{X^{\theta} \leqslant |p|^2 < X^{1/2}} S(\mathcal{E}_p, p),$$

and $S(\mathcal{E}_p, p) = S'(\mathcal{E}_p, p)$. For $p \in \Pi_1$, the second identity of (5), with $q = p$ and $w = z(|p|^2) \leqslant |p|^2$, gives

$$S'(\mathcal{E}_p, p) = S\big(\mathcal{E}_p, z(|p|^2)\big) - \sum_{\substack{q \prec p \\ |q|^2 \geqslant z(|p|^2)}} S(\mathcal{E}_{pq}, q).$$

Together they give (8).

The term $S(\mathcal{E}, X^{\theta})$ of (8) has an asymptotic formula by Corollary 6. In the sum of $S(\mathcal{E}_p, z(|p|^2))$ over $p \in \Pi_1$, the level $z(|p|^2)$ does not exceed $|p|^2$, so $S = S'$. For $|p|^2 \geqslant W$ the level is $X^{\tau}/|p|^2$ and $|p|^2 < X^{1/2} = X^{\tau-\theta}$, so Lemma 4 applies. For $|p|^2 < W$, Lemma 5 applies with the level $z(|p|^2)$.

In the sum over $p \in \Pi_2$, the term $S(\mathcal{E}_p, p)$ differs from $S'(\mathcal{E}_p, |p|^2)$ only by the cofactors divisible by the other prime p' of norm $|p|^2$, when it exists and $p' \prec p$. So $0 \leqslant S'(\mathcal{E}_p, |p|^2) - S(\mathcal{E}_p, p) \leqslant |\mathcal{E}_{pp'}|$. For $p \in \Pi_2$, $|p|^2 < z(|p|^2)$ and $|p|^2 < W$, so Lemma 5 applies with the level $|p|^2$. Since $|pp'|^2 = |p|^4 < X^{\tau/2}$, Lemma 1(a) gives

$$\sum_p |\mathcal{A}_{pp'}| = \frac{2}{\pi W} \sum_p |\mathcal{B}_{pp'}| + \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big), \qquad \sum_p |\mathcal{B}_{pp'}| \ll X \sum_{|p|^2 \geqslant X^{\theta}} |p|^{-4} \ll X^{1-\theta}.$$

Hence the sum over $p \in \Pi_2$ has an asymptotic formula.

By the definition of β_0, a prime p with $|p|^2 = X^{\alpha}$ and $\theta \leqslant \alpha < 1/2$ lies in Π_2 exactly when $\beta_0(\alpha) > \alpha$. For $\alpha \geqslant \sigma$, $\beta_0(\alpha) = \tau - \alpha \leqslant \alpha$. For $\alpha < \sigma$ and $j = j(X^{\alpha})$, the inequality $\beta_0(\alpha) > \alpha$ means $\alpha < \tau/(j+3)$. For $j = 0$ this is impossible, because $j = 0$ gives $\alpha \geqslant \sigma - \theta > \tau/3$. For $j = 1$ it gives $\alpha \in [\sigma - 2\theta, \tau/4)$, and $\tau/4 \leqslant \sigma - 2\theta$ for $\theta \leqslant 3/26$. For $j = 2$ it gives $\alpha \in [\max(\theta, \sigma - 3\theta), \tau/5)$. For $j \geqslant 3$, $\alpha < \sigma - 3\theta < \theta$, which is excluded. □

At $\theta = 0.1129$ the primes of Π_2 have $\alpha \in [0.1129, 0.12258)$. Since $\sum^*$ runs over $p \in \Pi_1$ only, the primes of Π_2 enter none of the discarded sums below.

In the sum $\sum^*$, a term with $|p|^2|q|^4 \geqslant X$ has cofactor $t = 1$, because $t \neq 1$ would give $|pqt|^2 \geqslant |p|^2|q|^4 \geqslant X$. Then $pq \in \mathcal{E}$ and $q \prec p$ force $|p|^4 \geqslant X/2$ and $|q|^2 \geqslant X^{1/2}/2$, and the number of such pairs with $pq \in \mathcal{B}$ is $\mathcal{O}(X/(\log X)^2)$. These terms are non-negative and enter (8) with the sign +, so we discard them. Their contribution tends to 0 after division by $S(\mathcal{B}, X^{1/2})$, which is of order $X/\log X$ by (26). The remaining pairs have exponents in the domain D of (9).

On each part of Ranges 1 to 6, the hypotheses of the lemmas used there are linear inequalities in α, β and θ. Appendix D lists the inequalities between the parameters that these hypotheses use, with their values at $\theta = 0.1129$.

Range 1 consists of the pairs with $\sigma \leqslant \alpha < 1/2$. Here $\beta_0(\alpha) = \tau - \alpha$ and $\beta < (1-\alpha)/2 < \alpha$. Lemma 9 gives an asymptotic formula where $\alpha \leqslant \zeta$, $\beta \geqslant \alpha - 2\theta$ and $\alpha + \beta \leqslant \nu$. The discarded region is

$$\mathsf{D}_1 = \big\{\sigma \leqslant \alpha < 1/2,\ \tau - \alpha \leqslant \beta < (1-\alpha)/2\big\} \setminus \big\{\alpha - 2\theta \leqslant \beta \leqslant \nu - \alpha\big\}.$$

The set subtracted in D_1 is empty for $\alpha > \chi$, where $\chi = (\nu + 2\theta)/2$ is the exponent in the hypothesis $P \ll X^{\chi}$ of Lemma 11 of Harman and Lewis [7] (Appendix E). Since $\chi < \zeta$, the condition $\alpha \leqslant \zeta$ of Lemma 9 holds wherever this set is non-empty.

Range 2 consists of the pairs with $1/3 \leqslant \alpha < \sigma$. Here $j(X^{\alpha}) = 0$, so $\beta_0(\alpha) = (\tau - \alpha)/2$, and $\beta < (1-\alpha)/2$. Lemma 11 applies in case (a) where $\alpha + \beta \leqslant \zeta$, and in case (b) on the region R of (10). In both cases $\beta \geqslant (\tau - \alpha)/2$, so $S_< = 0$, and the third sum of (18) is discarded. The two-dimensional discarded region is

$$\mathsf{D}_2 = \big\{1/3 \leqslant \alpha < \sigma,\ \max(\zeta - \alpha, (\tau - \alpha)/2) \leqslant \beta < (1-\alpha)/2,\ \beta > \rho/2\big\}.$$

Range 3 consists of the pairs with $\rho \leqslant \alpha < 1/3$. Here $\beta_0(\alpha) = (\tau - \alpha)/2$ and $\beta < \alpha$. Lemma 11 applies in case (a) where $\alpha + \beta \leqslant \zeta$, and the discarded region is

$$\mathsf{D}_4 = \big\{\rho \leqslant \alpha < 1/3,\ \zeta - \alpha \leqslant \beta < \alpha\big\}.$$

Range 4 consists of the pairs with $\sigma-\theta\leqslant\alpha<\rho$. Here $\beta_0(\alpha)=(\tau-\alpha)/2$. Lemma 11 applies in case (a) where $\alpha+2\beta\leqslant\nu$, and the discarded region is

$$\mathsf{D}_6=\big\{\sigma-\theta\leqslant\alpha<\rho,\ (\nu-\alpha)/2\leqslant\beta<\alpha\big\}.$$

Range 5 consists of the pairs with $\alpha<\sigma-\theta$ and $\alpha+\beta\geqslant\sigma-\theta$. Where $\alpha+\beta\geqslant\sigma$ and $\beta<(\tau-\alpha)/2$, the sum equals $S_<$ of Lemma 11 and has an asymptotic formula. Where $\alpha+\beta\geqslant\sigma$, $\beta\geqslant(\tau-\alpha)/2$ and $\alpha+2\beta\leqslant\nu$, Lemma 11 applies in case (a). Where $\alpha+\beta<\sigma$, Lemma 12 with $j=0$ applies. The discarded region is

$$\mathsf{D}_8=\big\{\nu/3\leqslant\alpha<\sigma-\theta,\ (\nu-\alpha)/2\leqslant\beta<\alpha\big\}.$$

Range 6 consists of the pairs with $\alpha+\beta<\sigma-\theta$. Lemma 12 with $j=1$ applies. Here $\beta_0(\alpha)\geqslant(\tau-\alpha)/4$, and $\beta\geqslant(\tau-\alpha)/4$ is equivalent to $|q|^2\geqslant z(|pq|^2)$, so $S_<=0$ on every box at distance at least ε from the boundary of D.

In the third sum of (18), the exponents satisfy $\tau-\alpha-\beta\leqslant\delta<\gamma<\beta$. In that of (19), they satisfy

$$\frac{\tau-\alpha-\beta}{j+2}\leqslant\gamma<\beta,\qquad\frac{\tau-\alpha-\beta-\gamma}{j+1}\leqslant\delta<\gamma.$$

With these limits, the third sums are discarded over the following regions of $(\alpha,\beta,\gamma,\delta)$:

$$\begin{aligned}
&\mathsf{E}_3:1/3\leqslant\alpha<\sigma,\ (\tau-\alpha)/2\leqslant\beta\leqslant\zeta-\alpha,\ \tau-\alpha-\beta\leqslant\delta<\gamma<\beta;\\
&\mathsf{E}_\mathsf{R}:(\alpha,\beta)\in\mathsf{R},\ \tau-\alpha-\beta\leqslant\delta<\gamma<\beta;\\
&\mathsf{E}_5:\rho\leqslant\alpha<1/3,\ (\tau-\alpha)/2\leqslant\beta\leqslant\zeta-\alpha,\ \tau-\alpha-\beta\leqslant\delta<\gamma<\beta;\\
&\mathsf{E}_7:\sigma-\theta\leqslant\alpha<\rho,\ (\tau-\alpha)/2\leqslant\beta\leqslant(\nu-\alpha)/2,\ \tau-\alpha-\beta\leqslant\delta<\gamma<\beta;\\
&\mathsf{E}_9:\alpha<\sigma-\theta,\ (\tau-\alpha)/2\leqslant\beta\leqslant\min((\nu-\alpha)/2,\alpha),\ \tau-\alpha-\beta\leqslant\delta<\gamma<\beta;\\
&\mathsf{E}_{10}:\tau/4\leqslant\alpha\leqslant\sigma-\theta,\ (\tau-\alpha)/3\leqslant\beta<\min(\alpha,\sigma-\alpha),\ (\tau-\alpha-\beta)/2\leqslant\gamma<\beta,\ \tau-\alpha-\beta-\gamma\leqslant\delta<\gamma;\\
&\mathsf{E}_{11}:\tau/5\leqslant\alpha\leqslant\sigma-2\theta,\ (\tau-\alpha)/4\leqslant\beta<\min(\alpha,\sigma-\theta-\alpha),\ (\tau-\alpha-\beta)/3\leqslant\gamma<\beta,\ (\tau-\alpha-\beta-\gamma)/2\leqslant\delta<\gamma.
\end{aligned}\tag{23}$$

In E_{10}, $\beta\geqslant(\tau-\alpha)/3$ implies $\alpha+\beta\geqslant\tau/2>\sigma-\theta$, so the pairs of E_{10} lie in Range 5. The region E_{11} contains the set of exponents over which the third sum is discarded in Range 6, since $\beta_0(\alpha)\geqslant(\tau-\alpha)/4$. For $\theta<3/26$, the primes with $\alpha\in[\tau/4,\sigma-2\theta)$ have $j(X^\alpha)=2$ and $\beta_0(\alpha)=(\tau-\alpha)/4$. Their pairs lie in Ranges 5 and 6, where Lemma 12 does not depend on $j(X^\alpha)$, and the discarded third sums of these pairs lie in E_{10} and E_{11}.

Lemmas 14 and 16 need a positive lower bound for the exponents. All exponents on D and on the regions (23) are at least

$$c(\theta)=\min\Big(4\theta^2,\ \frac{5\theta}{2}-\frac14,\ \frac{26\theta-1}{24},\ \frac{1-2\theta}{6}\Big)>0.\tag{24}$$

On D, $\alpha\geqslant\theta$ and $\beta\geqslant\beta_0(\alpha)$. For $\alpha\geqslant\sigma$, $\beta_0(\alpha)\geqslant\theta$. For $\alpha<\sigma$, $\beta_0(\alpha)=(\tau-\alpha)/(j+2)>2\theta/(j+2)>4\theta^2$, because $j+2<(\sigma+\theta)/\theta=1/(2\theta)$. On the regions (23), $\alpha\geqslant\beta\geqslant\gamma\geqslant\delta$. Where $\alpha+\beta\leqslant\zeta$ or $(\alpha,\beta)\in\mathsf{R}$, $\delta\geqslant\tau-\alpha-\beta\geqslant\min((1-2\theta)/6,(26\theta-1)/24)$. On E_{10}, $\delta\geqslant\tau-3\sigma/2=5\theta/2-1/4$. On E_{11}, $\delta\geqslant2\theta-1/8\geqslant5\theta/2-1/4$.

8 The four-dimensional sums

This section proves that the third sums of (18) and (19) have asymptotic formulas on the regions G and L of (11). Let G_ε and L_ε be the regions defined by the same inequalities with margin ε.

Lemma 14 *Let $p\sim P$, $q\sim Q$ be a box on which Lemma 11 or Lemma 12 gives identity (18) or (19), and split the third sum of the identity into the sums over $r\sim R$ and $s\sim R'$, with R,R' powers of 2. The quadruples (p,q,r,s) with $p\sim P$, $q\sim Q$, $r\sim R$ and $s\sim R'$ form a box, whose* exponent box *is defined in the same way as the exponent rectangle of Section 6. If the exponent box lies in G_ε, or in L_ε, then the sum of $S(\mathcal{A}_{pqrs},s)$ over the box has an asymptotic formula.*

Proof. By (24), all exponents on the regions (23) are at least $c(\theta)$, so all ranges lie in $[\mu^4, X]$. The sum is the configuration with variables p, q, r, s, the box ranges, the constraints $q \prec p$, $r \prec q$ and $s \prec r$, the lower limits of r and s from the identity, and the sieve condition of type (S) with i_s the index of s. The lower limits are constants for (18). For (19), the lower limit of s is the monomial constraint $|r|^2|s|^{2(j+1)} \geqslant X^\tau/(PQ)$. By Lemma 7, it suffices to check the pieces.

Suppose the exponent box lies in G_ε, and let $\mathcal{S} \subset \{p, q, r\}$ be a non-empty set with $\sigma \leqslant x_{\mathcal{S}} \leqslant \tau - \delta$, with margin ε, on the exponent box. Every piece of hypotheses (a), (b) and (c) falls under (P3) of Lemma 8, with this $\mathcal{S}$, with $\varpi_{i_0} = s$, and with $\mathcal{S}'$ the other primes, together with ϖ' for the window pieces. Indeed, the primes of $\mathcal{S}$ have product of norm at least $X^{\sigma+\varepsilon/2}$, and the quotient of (P3) is at least $X^{1-x_{\mathcal{S}}-\delta-\varepsilon/2} \geqslant X^{\sigma+\varepsilon/2}$.

Suppose the exponent box lies in L_ε. Buchstab's identity (5) at level μ^4, with s' running over primes, gives

$$S'(\mathcal{A}_{pqrs}, s) = S'(\mathcal{A}_{pqrs}, \mu^4) - \sum_{s' \prec s,\ |s'|^2 \geqslant \mu^4} S'(\mathcal{A}_{pqrss'}, s'). \tag{25}$$

Summed over the box, the term $S'(\mathcal{A}_{pqrs}, \mu^4)$ of (25) is the configuration with the same variables and constraints, and with the sieve condition of type (C). Its pieces of hypothesis (a) fall under (P2), and its straddling pieces under (P1), by the condition I with margin ε. The sum of $S'(\mathcal{A}_{pqrss'}, s')$ is the configuration with the further prime variable s', the constraint $s' \prec s$, and the sieve condition of type (S) with i_s the index of s'. Let $\mathcal{S} \subset \{p, q, r, s\}$ be a set with $\sigma \leqslant x_{\mathcal{S}} \leqslant \tau - \delta$, with margin ε, on the exponent box. Since $|s'|^2 \leqslant |s|^2 = X^\delta$, the quotient of (P3) is at least $X^{1-x_{\mathcal{S}}-\delta-\varepsilon/2} \geqslant X^{\sigma+\varepsilon/2}$. So every piece falls under (P3) with this $\mathcal{S}$, with $\varpi_{i_0} = s'$, and with $\mathcal{S}'$ the other primes, together with ϖ' for the window pieces. Both terms of (25) have asymptotic formulas, and so does their difference. □

The conditions of G and L use the smallest prime of the tuple, s or s', as the prime of the bilinear estimate. The same proof applies with any other prime of the tuple in this role. Say that a set $\mathcal{S}$ of primes of the tuple *works* with a prime outside $\mathcal{S}$ of exponent x if $\sigma \leqslant x_{\mathcal{S}} \leqslant \tau - x$. Let the smallest prime have exponent x_0, let another prime have exponent $x_1 \geqslant x_0$, and let $\mathcal{S}$ work with the prime of exponent x_1. If the smallest prime is not in $\mathcal{S}$, then $\mathcal{S}$ works with it, since $\tau - x_1 \leqslant \tau - x_0$. If it is in $\mathcal{S}$, replacing it by the prime of exponent x_1 gives a set that works with the smallest prime. So the choice of the smallest prime in (11) does not make the regions smaller.

9 The discarded sums and the lower bound

This section proves Lemma 16, the last step of Section 3. An element of $\mathcal{B}$ without prime factors of norm below $X^{1/2}$ is a prime, so Lemma 17(d) gives

$$S(\mathcal{B}, X^{1/2}) = \pi_G(X) - \pi_G(X/2) = (1 + o(1))\frac{X}{2\log X}. \tag{26}$$

The next lemma evaluates the discarded sums over $\mathcal{B}$.

Lemma 15 *Let $K \geqslant 1$ and $c_1 > 0$, and let $\mathcal{R} \subset \mathbb{R}^K$ be a bounded set whose boundary has Lebesgue measure 0, on which $y_i \geqslant c_1$ and $y_K \leqslant y_i$ for all i, and $\sum_i y_i \leqslant 1 - c_1$. With $y(\varpi) = (\log|\varpi_i|^2/\log X)_{i \leqslant K}$ for a tuple of primes $\varpi = (\varpi_1, \dots, \varpi_K)$,*

$$\sum_{\substack{y(\varpi) \in \mathcal{R} \\ \varpi_K \prec \varpi_i \ (i<K)}} S(\mathcal{B}_{\varpi_1 \cdots \varpi_K}, \varpi_K) = S(\mathcal{B}, X^{1/2})\Big(\int_{\mathcal{R}} \omega\Big(\frac{1 - \sum_i y_i}{y_K}\Big)\frac{dy}{y_1 \cdots y_{K-1} y_K^2} + o(1)\Big). \tag{27}$$

Appendix C proves Lemma 15. The lower bound below uses (27) only as an upper bound for discarded sums.

Lemma 16 *For every small $\varepsilon > 0$ there is a number $\Phi_\varepsilon(\theta)$, with $\Phi_\varepsilon(\theta) \to \Phi(\theta)$ as $\varepsilon \to 0$, such that*

$$S(\mathcal{A}, X^{1/2}) \geqslant \frac{2}{\pi W} S(\mathcal{B}, X^{1/2})\big(1 - \Phi_\varepsilon(\theta) - o(1)\big). \tag{28}$$

The same holds with Φ_{P} in place of Φ. In particular, if $\Phi(\theta) < 1$ or $\Phi_{\mathsf{P}}(\theta) < 1$, then $S(\mathcal{A}, X^{1/2}) \gg XW^{-1}/\log X$.

Proof. Identity (8) holds for $\mathcal{A}$ and for $\mathcal{B}$, and its terms other than $\sum^*$ have asymptotic formulas by Lemma 13. The domain D of (9) is divided by Ranges 1 to 6 of Section 7 into finitely many parts, which are polytopes. The *treatment* of a part is the one that Section 7 applies there: an asymptotic formula for the whole sum, identity (18) or (19), or a discard of the whole sum. The hypotheses of each lemma are finitely many linear inequalities in (α, β). They hold on the closed polytope where the lemma is applied (Appendix D). Hence they hold with margin proportional to ε at the points of this polytope at distance at least ε from its boundary.

Split $\sum^*$ into the $\mathcal{O}((\log X)^2)$ dyadic boxes $p \sim P$, $q \sim Q$. Each box has an exponent rectangle of side $\log 2/\log X$. Call a box *good* if its exponent rectangle lies inside a part whose treatment is not a discard, at distance at least ε from the boundary of that part. Then all its pairs belong to $\sum^*$, and the corresponding lemma applies with margin. All other boxes are discarded. This is legitimate because every term of $\sum^*$ is non-negative and enters (8) with the sign $+$. The same holds for the third sums of (18) and (19). For a discarded box $p \sim P$, $q \sim Q$, put

$$\Delta_{P,Q}(\mathcal{B}) = \sum_{p\sim P,\, q\sim Q}^{*} S(\mathcal{B}_{pq}, q).$$

For a good box, let $\Delta_{P,Q}(\mathcal{B})$ be the third sum of the identity used on the box if an identity is used, and 0 otherwise. Summing the $\mathcal{O}((\log X)^2)$ errors and using (8) for $\mathcal{B}$, we obtain

$$S(\mathcal{A}, X^{1/2}) \geqslant \frac{2}{\pi W}\Big(S(\mathcal{B}, X^{1/2}) - \sum_{P,Q} \Delta_{P,Q}(\mathcal{B})\Big) - \mathcal{O}\big(XW^{-1}(\log X)^{2-A}\big). \tag{29}$$

In the third sums, split r and s into dyadic ranges as in Lemma 14. Call such a four-dimensional box *good* if its exponent box lies inside G_ε, or inside L_ε, at distance at least ε from the boundary of that set. Lemma 14 gives every good four-dimensional box an asymptotic formula, so it can be moved from the discarded terms to the main term. Every other four-dimensional box, including a box that meets both sets but lies in neither at distance at least ε from its boundary, stays discarded. The number of four-dimensional boxes is $\mathcal{O}((\log X)^4)$, and the errors of the good ones add up to $\mathcal{O}(XW^{-1}(\log X)^{4-A})$.

A discarded box $p \sim P$, $q \sim Q$ has its exponent rectangle within distance 2ε of a part whose treatment is a discard, of a boundary between parts, or of $\partial\mathsf{D}$. A discarded box $p \sim P$, $q \sim Q$, $r \sim R$, $s \sim R'$ has its exponent box within distance 2ε of the set of points of the regions (23) outside $\mathsf{G} \cup \mathsf{L}$, or of the boundary of G or of L. The third sums use the quantities $X^\tau/(PQ)$ and $z(PQ)$, which are constant on a box, in place of $X^\tau/|pq|^2$ and $z(|pq|^2)$. This moves the lower limits of γ and δ in (23) by $\mathcal{O}(1/\log X)$. Let $\Phi_\varepsilon(\theta)$ be the sum of the losses (13) of the 2ε-neighborhoods of these sets, intersected with the closures of D and of the regions (23). Lemma 15, with $K = 2$ and $K = 4$, bounds the discarded sums over $\mathcal{B}$ by $S(\mathcal{B}, X^{1/2})(\Phi_\varepsilon(\theta) + o(1))$. Its hypothesis $y_i \geqslant c_1$ holds with $c_1 = c(\theta)$ by (24).

As $\varepsilon \to 0$, these neighborhoods decrease to the closures of the regions in (14), together with finitely many boundaries. The sets G, L and P are finite unions of polytopes, so these boundaries are null sets. The integrands are bounded, so $\Phi_\varepsilon(\theta) \to \Phi(\theta)$. Finally $XW^{-1}(\log X)^{4-A} = o(W^{-1}S(\mathcal{B}, X^{1/2}))$ for $A > 5$. Inserting these bounds into (29) gives (28). The same argument applies with P in place of L. □

10 Evaluation of the losses and proof of Theorem 1

The losses (13) are computed numerically by two independent methods, implemented in the folder `Code` (see the data availability statement). Buchstab's function ω and its integral $\Omega(u) = \int_1^u \omega(t)\,dt$ are tabulated on a grid of step 10^{-5} from the relation that defines ω below (13). On $[1, 3]$, the tables agree with the closed forms of ω and Ω to 10^{-14}.

In the first method, the innermost integral is evaluated exactly through Ω: over β in the two-dimensional regions, and over the part of the δ-range outside $\mathsf{G}\cup\mathsf{L}$ in the four-dimensional regions, which is a finite union of intervals. The outer integrals are computed by composite Gauss–Legendre quadrature. For the regions D_k, the α-axis is cut into 256 equal panels. For the regions E_k, each of the α-, β- and γ-axes is cut into 16 equal panels. The panels are cut further at the points where a limit or the set $\mathsf{G} \cup \mathsf{L}$ changes its formula, and each resulting subinterval carries 10 Gauss–Legendre nodes.

In the second method, the two-dimensional losses are computed by nested adaptive quadrature of the integrand itself [10]. The four-dimensional losses are computed by scrambled Sobol quasi-Monte Carlo

integration [9, 11], with ω evaluated directly and with the sets G and L tested pointwise. Each of these losses is the mean of 8 estimates, from independent scramblings of 2^{20} points each.

Table 2 lists the losses at $\theta = 0.1129$. The first method gives $\Phi(0.1129) = 0.9990033$, and the second gives 0.9990044. At $\theta = 0.119$, the decomposition of Harman and Lewis, with the regions derived from the lemmas (Remark 27), gives the total 0.9425585, which matches the bound 0.943 of Harman and Lewis [7, p. 134]. Table 3 lists the values of θ at which the total loss equals 1 as the groupings are added.

Region	First method	Second method
D_1	0.3201605	0.3201605
D_2	0.3437028	0.3437028
D_4	0.2132922	0.2132922
D_6	0.0480512	0.0480512
D_8	0.0684437	0.0684437
$\mathsf{E}_3 \setminus (\mathsf{G} \cup \mathsf{L})$	0.0000098	0.0000098
$\mathsf{E}_\mathsf{R} \setminus (\mathsf{G} \cup \mathsf{L})$	0.0010206	0.0010207
$\mathsf{E}_5 \setminus (\mathsf{G} \cup \mathsf{L})$	0.0006199	0.0006199
$\mathsf{E}_7 \setminus (\mathsf{G} \cup \mathsf{L})$	0.0004451	0.0004451
$\mathsf{E}_9 \setminus (\mathsf{G} \cup \mathsf{L})$	0.0002727	0.0002726
$\mathsf{E}_{10} \setminus (\mathsf{G} \cup \mathsf{L})$	0.0028858	0.0028870
$\mathsf{E}_{11} \setminus (\mathsf{G} \cup \mathsf{L})$	0.0000988	0.0000988
$\Phi(0.1129)$	0.9990033	0.9990044

Table 2: The losses (13) at $\theta = 0.1129$ of the regions of Table 1, with $\mathsf{G} \cup \mathsf{L}$ removed from the regions E_k, computed by the methods of Section 10 and rounded to 7 decimal places.

Decomposition	θ with total loss 1	$1/2 - 2\theta$
Harman and Lewis	0.1167825	0.2664351
with R	0.1141073	0.2717854
with R and G	0.1139312	0.2721376
with R, G and P	0.1132049	0.2735902
with R, G and L	0.1128554	0.2742892

Table 3: The value of θ at which the total loss equals 1, for the decomposition of Harman and Lewis with the regions derived from the lemmas (Remark 27), and with the groupings listed in each row added to it, computed by the first method of Section 10. The fourth row uses the polytope P in place of L. The last column is the exponent $1/2 - 2\theta$ in $\{\sqrt{p}\} < p^{-(1/2-2\theta)}$.

The computation is not carried out in interval arithmetic. For the first method, the total at $\theta = 0.1129$ changes by 2.6×10^{-10} between the rule above and the finest rule tested, which has 512 panels on the α-axis for the regions D_k, 32 panels per axis for the regions E_k, and 12 nodes per subinterval. For each region E_k, the error estimate of the second method is the square root of the sum of the squared deviations of its 8 estimates from their mean, divided by $(8 \cdot 7)^{1/2}$. Over the regions E_k, the square root of the sum of the squares of these error estimates is 1.0×10^{-6}. The methods differ by 1.2×10^{-6}, which is 1.1 times this value. The margin $1 - \Phi(0.1129) = 9.97 \times 10^{-4}$ exceeds each of these quantities by a factor of more than 800.

Proof of Theorem 1. Let $\theta = 0.1129$. By Table 2, $\Phi(\theta) < 1$, so Lemma 16 gives $S(\mathcal{A}, X^{1/2}) \gg XW^{-1}/\log X = X^{1/2+\theta}/\log X$, uniformly in ψ. Since $S(\mathcal{A}, X^{1/2})$ is the number of Gaussian primes in $\mathcal{A}$, this is Theorem 1. □

For the polytope P of (12), the methods give $\Phi_\mathsf{P}(0.1133) = 0.9978243$ and 0.9978250. With P in place of L, the same argument gives Theorem 1 with $\theta = 0.1133$, and hence $\{\sqrt{p}\} < p^{-0.2734}$ for infinitely many primes.

11 Conclusion

Of the total loss 0.999 at $\theta = 0.1129$, the regions D_1, D_2 and D_4 account for 0.877. On these regions, no factorization of pqr satisfies the conditions of the linear estimate once $|r|^2$ is close to $|q|^2$. We have not found another grouping that gives them asymptotic formulas. A further improvement may need new arithmetic information for the Hecke characters λ^m of Appendix A. Harman and Lewis [7, p. 134] call the improvement of their arithmetic information "the more interesting challenge". The numerical evaluation of the losses could also be made rigorous by interval arithmetic.

Data availability

The code that computes every number in Section 10 and Appendix D is in the folder `Code` of the accompanying repository, with a description of each computation and its output.

A The linear and bilinear estimates

This appendix derives Lemmas 1 and 2 of Section 4 from the arithmetic information. It begins with standard facts, which Sections 5 and 9 and Appendices B and C also use. Lemma 17 bounds sums over Gaussian primes, and Lemma 18 counts Gaussian integers and their divisors. Lemma 19 is the truncated Perron formula used in the proof of Lemma 2.

Lemma 17 *We have $\pi_G(y) = \mathrm{Li}(y) + \mathcal{O}(y\exp(-c(\log y)^{1/2}))$ for some $c > 0$, where $\mathrm{Li}(y) = \int_2^y dt/\log t$. Consequently, for fixed $B \geqslant 0$ and $\kappa_0 \geqslant 1$ and with $H = (\log X)^B$, the following hold once X is large.*

(a) If $Y \geqslant \mu$ and $1 \leqslant \kappa \leqslant \kappa_0$, then

$$\sum_{Y \leqslant |\varpi|^2 < Y(1+1/H)^\kappa} |\varpi|^{-2} \leqslant \frac{2\kappa}{H \log Y}.$$

(b) $\sum_{|\varpi|^2 < y} |\varpi|^{-2} \leqslant \log\log y + C$ for $y \geqslant 3$.

(c) For fixed $c_1 > 0$, uniformly for $c_1 \leqslant a < b \leqslant 1$,

$$\sum_{X^a \leqslant |\varpi|^2 < X^b} |\varpi|^{-2} = \log\frac{b}{a} + \mathcal{O}\Big(\frac{1}{\log X}\Big).$$

(d) $\pi_G(y) - \pi_G(y/2) = (1 + \mathcal{O}(1/\log y))\, y/(2\log y)$.

Proof. For $a \in \{1,3\}$, let $\pi(y;4,a)$ be the number of rational primes $\ell < y$ with $\ell \equiv a \pmod 4$. A Gaussian prime is $1+i$, a prime of norm ℓ for a rational prime $\ell \equiv 1 \pmod 4$, or a rational prime $\ell \equiv 3 \pmod 4$, of norm ℓ^2. Hence $\pi_G(y) = 2\pi(y;4,1) + \pi(y^{1/2};4,3) + \mathcal{O}(1)$. The first statement follows by partial summation from the prime number theorem in progressions modulo 4 [12, Section 1.2], in the form

$$\sum_{\ell \leqslant x,\ \ell \equiv a\,(4)} \log\ell = \frac{x}{2}\Big(1 + \mathcal{O}\big(\exp(-c'(\log x)^{3/5}(\log\log x)^{-1/5})\big)\Big), \qquad a \in \{1,3\},$$

with a constant $c' > 0$. Statements (b), (c) and (d) follow by partial summation. In (a), the main term contributes at most $(\kappa+1)/(H\log Y)$. The error term contributes $\mathcal{O}(\exp(-c(\log Y)^{1/2}))$. Since $\log Y \geqslant (\log X)^{4/5}$, this is $\mathcal{O}(\exp(-c(\log X)^{2/5}))$, which is smaller than $1/(H\log X)$. □

Lemma 18 *(a) For $M \geqslant 1$, the number of Gaussian integers u with $X/(2M) \leqslant |u|^2 < X/M$ is at most CX/M. In particular $|\mathcal{B}_u| \leqslant CX/|u|^2$.*

(b) For $k \geqslant 1$ and $Y \geqslant 2$,

$$\sum_{|n|^2 \leqslant Y} d(n)^k \ll_k Y(\log Y)^{2^k}, \qquad \sum_{|n|^2 \leqslant Y} d(n)^k |n|^{-2} \ll_k (\log Y)^{2^k}.$$

(c) A Gaussian integer n has at most $d(n)^{h-1}$ ordered factorizations into h factors.

Proof. For (a), the count is 0 if $X/M < 1$, and $\pi X/(8M) + \mathcal{O}((X/M)^{1/2} + 1)$ otherwise. For (b), d is multiplicative with $d(\varpi^e) = e + 1$. So, by Lemma 17(b), the sum of $d(n)^k |n|^{-2}$ is at most

$$\prod_{|\varpi|^2 \leqslant Y} \sum_{e \geqslant 0} (e+1)^k |\varpi|^{-2e} \leqslant \exp\Big(\sum_{|\varpi|^2 \leqslant Y} \big(2^k |\varpi|^{-2} + C_k |\varpi|^{-4}\big) \Big) \ll_k (\log Y)^{2^k},$$

and the sum of $d(n)^k$ is at most Y times the sum of $d(n)^k |n|^{-2}$. For (c), the number $d_h(n)$ of ordered factorizations satisfies $d_h(n) = \sum_{u|n} d_{h-1}(n/u) \leqslant d(n)^{h-1}$, by induction on h. □

Lemma 19 *For $y > 0$, $c > 0$ and $T_0 \geqslant c$, let*

$$I(y) = \frac{1}{2\pi i} \int_{c - iT_0}^{c + iT_0} y^s \, \frac{ds}{s}.$$

Then $|I(y)| \leqslant y^c \log(3T_0/c)$, and $|I(y) - 1[y > 1]| \leqslant y^c/(\pi T_0 |\log y|)$ for $y \neq 1$.

Proof. The first bound follows from $|I(y)| \leqslant (y^c/2\pi) \int_{-T_0}^{T_0} (c^2 + t^2)^{-1/2} \, dt$. For $y > 1$, move the segment to $\operatorname{Re} s = -K$ and let $K \to \infty$. The residue at 0 is 1, the integral over $\operatorname{Re} s = -K$ tends to 0, and each horizontal side is at most

$$\frac{1}{2\pi T_0} \int_{-\infty}^{c} y^u \, du = \frac{y^c}{2\pi T_0 \log y}.$$

For $y < 1$, move the segment to $\operatorname{Re} s = K$, where there is no pole, and obtain the same bound with $|\log y|$. □

For an integer m and a Gaussian integer $n \neq 0$ put $\lambda^m(n) = (n/|n|)^{4m}$, as in Harman and Lewis [7, p. 122]. This is a Hecke character. It is unchanged when n is multiplied by a unit, so it is completely multiplicative on the representatives. Since $\lambda^m(n) = e^{4im \arg n}$, it detects the argument of n modulo $\pi/2$. The following results are quoted without proof: Harman and Lewis [7, (12), Lemmas 1 and 4], and their Lemmas 6 and 7, which are combined into a single statement. Lemmas 20 and 23 give the linear estimate, Lemma 1. Lemmas 20, 21 and 22 give the bilinear estimate, Lemma 2.

Lemma 20 (Harman and Lewis [7], (12)) *For every finite set $\mathcal{E}$ of Gaussian integers, every $T \geqslant 1$ and all weights a_n,*

$$\sum_{\substack{n \in \mathcal{E} \\ \psi \leqslant \arg n < \psi + W^{-1}}} a_n = \frac{2}{\pi W} \sum_{n \in \mathcal{E}} a_n + \mathcal{O}\Big(\frac{1}{T} \sum_{n \in \mathcal{E}} |a_n| \Big) + \mathcal{O}\Big(\frac{1}{W} \sum_{m=1}^{T} \Big| \sum_{n \in \mathcal{E}} a_n \lambda^m(n) \Big| \Big).$$

Harman and Lewis derive Lemma 20 from Baker [2, Chapter 2]. We apply it only with non-negative weights, and with $T = W(\log X)^A$, so that the first error term is $W^{-1}(\log X)^{-A} \sum |a_n|$.

Lemma 21 (Harman and Lewis [7], Lemma 1) *For $T \geqslant 1$, $N \geqslant 1$ and all coefficients $c(n)$,*

$$\sum_{m=1}^{T} \Big| \sum_{|n|^2 \leqslant N} c(n) \lambda^m(n) \Big|^2 \ll (T + N) \sum_{|n|^2 \leqslant N}{}' |c_1(n)|^2, \qquad c_1(n) = \sum_{\arg u = \arg n} c(u),$$

where $\sum'$ runs over the primitive Gaussian integers $n = x + iy$, $\gcd(x, y) = 1$.

Lemma 22 (Harman and Lewis [7], Lemma 4) *Let $R, U \geqslant 2$, $T \geqslant S \geqslant R + \exp((\log TU)^{4/5})$, $1 \leqslant m \leqslant T$ and $|t| \leqslant U$. Then*

$$\Big| \sum_{R \leqslant |\varpi|^2 < S} |\varpi|^{2it} \lambda^m(\varpi) \Big| \ll S \exp\Big(-\frac{\log S}{2(\log TU)^{7/10}} \Big).$$

Lemma 21 is Coleman [5, Lemma 6], and Lemma 22 rests on the zero-free region of Coleman [4].

Lemma 23 (Harman and Lewis [7], Lemmas 6 and 7) *Let $A > 0$, $T = W(\log X)^A$, and let a_u, b_v be divisor-bounded.*

(a) If $U \leqslant X^\tau \mu^{-1}$, then

$$\sum_{m=1}^{T} \Big| \sum_{u \sim U} a_u \sum_{uv \in \mathcal{B}} \lambda^m(uv) \Big| \ll X(\log X)^{-A}.$$

(b) If $U \leqslant X^\zeta \mu^{-1}$, $V \leqslant X^\rho \mu^{-1}$ and $UV \leqslant X^\nu \mu^{-1}$, then

$$\sum_{m=1}^{T} \Big| \sum_{u \sim U} a_u \sum_{v \sim V} b_v \sum_{uvt \in \mathcal{B}} \lambda^m(uvt) \Big| \ll X(\log X)^{-A}.$$

Here v in (a) and t in (b) run over all Gaussian integers with the stated product in $\mathcal{B}$.

Remark 24 in Appendix E explains the conditions $uv \in \mathcal{B}$ and $uvt \in \mathcal{B}$ of Lemma 23.

Proof of Lemma 1. With $c'_n = \sum_{v|n} c_v \geqslant 0$,

$$\sum_v c_v |\mathcal{A}_v| = \sum_{n \in \mathcal{A}} c'_n, \qquad \sum_{n \in \mathcal{B}} c'_n = \sum_v c_v |\mathcal{B}_v|.$$

Apply Lemma 20 to $\mathcal{B}$ with $T = W(\log X)^A$. By Lemma 18, the first error term in Lemma 20 is at most

$$W^{-1}(\log X)^{-A} \cdot CX \sum_v c_v |v|^{-2} \ll XW^{-1}(\log X)^{C-A}.$$

The second error term in Lemma 20 is

$$\frac{1}{W} \sum_{m \leqslant T} \Big| \sum_v c_v \sum_{vt \in \mathcal{B}} \lambda^m(vt) \Big|.$$

Under (a), after the range of v is split into $\mathcal{O}(\log X)$ dyadic ranges, it is $\ll XW^{-1}(\log X)^{1-A}$ by Lemma 23(a). Under (b) it is $\ll XW^{-1}(\log X)^{-A}$ by Lemma 23(b). Since A is arbitrary, the lemma follows. □

Proof of Lemma 2. Fix A, put $T = W(\log X)^A$, and let E_m be the sum of $a_u b_v \lambda^m(uv\varpi)$ over the triples counted in (15) with $\mathcal{E} = \mathcal{B}$. As in the proof of Lemma 1, it suffices to show that $\sum_{m \leqslant T} |E_m| \ll X(\log X)^{-A}$. A triple occurs only if $X/(8UV) \leqslant |\varpi|^2 < X/(UV)$. Put $Z'_1 = \max(Z_1, X/(8UV))$ and $Z'_2 = \min(Z_2, X/(UV))$, and assume that $Z'_1 < Z'_2$. Let $c = 1/\log X$ and $T_0 = X^3$. For $N = |uv\varpi|^2 \leqslant 4X$ and $Y \in \{X, X/2\}$, apply Lemma 19 to $y = Y/N$. If $|N - Y| \geqslant 1/2$, then $|\log y| \geqslant 1/(16X)$ and the error is at most $16eX/(\pi T_0)$. At most 2 integers N have $|N - Y| < 1/2$, and for them we use $|I(y)| \leqslant e \log(3T_0/c)$. By Lemma 18(c), each N is the norm of $X^{o(1)}$ triples, each with weight $a_u b_v \leqslant X^{o(1)}$. Since λ^m is completely multiplicative, this gives

$$E_m = \frac{1}{2\pi i} \int_{c - iT_0}^{c + iT_0} \big(X^s - (X/2)^s \big) F_m(s) G_m(s) \mathfrak{p}_m(s) \, \frac{ds}{s} + \mathcal{O}(X^{o(1)}), \tag{30}$$

where

$$F_m(s) = \sum_{u \sim U} a_u \lambda^m(u) |u|^{-2s}, \qquad G_m(s) = \sum_{v \sim V} b_v \lambda^m(v) |v|^{-2s}, \qquad \mathfrak{p}_m(s) = \sum_{Z'_1 \leqslant |\varpi|^2 < Z'_2} \lambda^m(\varpi) |\varpi|^{-2s}.$$

Since $|X^s| = e$ and $\int_{-T_0}^{T_0} |c+it|^{-1}\,dt \ll \log X$, summing (30) over $m \leqslant T$ gives

$$\sum_{m\leqslant T} |E_m| \ll \log X \cdot \sup_{|t|\leqslant T_0} \sum_{m\leqslant T} \big|F_m G_m \mathfrak{p}_m\big|(c+it) + X^{\sigma+o(1)}. \tag{31}$$

By Cauchy's inequality, the sum over m is at most $\max_m |\mathfrak{p}_m|\,(\sum_m |F_m|^2)^{1/2}(\sum_m |G_m|^2)^{1/2}$.

Apply Lemma 21 with $2U$ in place of N and with $c(u) = a_u|u|^{-2s}$ for $u \sim U$. A Gaussian integer u with $\arg u = \arg n$, n primitive, is $u = kn$ with k a positive rational integer. So $|c_1(n)| \leqslant \sum_k d(kn)^{C_0}$, where $k \geqslant 1$ runs over the integers with $k^2|n|^2 \in [U, 2U)$. This sum has at most $(2U)^{1/2}/|n|$ terms. By Cauchy's inequality and $d(kn) \leqslant d(k)d(n)$,

$$\sum_n{}' |c_1(n)|^2 \leqslant (2U)^{1/2} \sum_{k^2\leqslant 2U} d(k)^{2C_0} \sum_{|n|^2<2U/k^2} d(n)^{2C_0}|n|^{-1}. \tag{32}$$

By Lemma 18(b) and partial summation, the sum of $d(n)^{2C_0}|n|^{-1}$ in (32) is $\ll (U/k^2)^{1/2}(\log X)^C$. The remaining sum over the rational integers $k \leqslant (2U)^{1/2}$, with weight $1/k$, is at most a product over the rational primes $\ell \leqslant (2U)^{1/2}$:

$$\prod_{\ell\leqslant (2U)^{1/2}} \sum_{e\geqslant 0} (e+1)^{4C_0}\ell^{-e} \ll (\log X)^{2^{4C_0}}.$$

Here $d(\ell^e) \leqslant (e+1)^2$ for every rational prime ℓ, and $\sum_{\ell\leqslant x} 1/\ell \leqslant \log\log x + \mathcal{O}(1)$ follows from the prime number theorem modulo 4. So the right side of (32) is $\ll U(\log X)^C$, and, since $T \leqslant C_1 U(\log X)^A$,

$$\sum_{m\leqslant T} |F_m|^2 \ll (T+U)U(\log X)^C \ll U^2(\log X)^{A+C}.$$

The same bound holds for G_m with V in place of U.

The weight $|\varpi|^{-2c}$ is decreasing in $|\varpi|^2$ and lies in $[e^{-2}, 1]$. By partial summation, $|\mathfrak{p}_m(c+it)|$ is at most 2 times the largest value, over $Z \in [Z_1', Z_2']$, of

$$\Big|\sum_{Z_1'\leqslant |\varpi|^2<Z} \lambda^m(\varpi)|\varpi|^{-2it}\Big|.$$

Each of these sums is the difference of the sums over $2 \leqslant |\varpi|^2 < S$ for $S = Z$ and for $S = Z_1'$. Lemma 22 applies to them with $R = 2$, with $|t| \leqslant T_0$, and with $1 \leqslant m \leqslant T$. Indeed, $S \leqslant X/(UV) \leqslant C_1^2 X^{2\theta} \leqslant T$ because $\theta < 1/6$. The bounds $\log(T_0T) \leqslant 4\log X$ and $4^{4/5} < 3.1$ give

$$S \geqslant \frac{\mu^4}{2} > 2 + \exp\big((\log T_0T)^{4/5}\big).$$

Since $S\exp(-\log S/(2(\log T_0T)^{7/10}))$ increases with S and $\log Z_2' \geqslant \log(\mu^4/2)$, Lemma 22 gives

$$\max_m |\mathfrak{p}_m| \ll \frac{X}{UV}\exp\Big(-\frac{3}{4}(\log X)^{1/10}\Big).$$

Collecting the bounds in (31), we obtain

$$\sum_{m\leqslant T} |E_m| \ll X(\log X)^{A+C+1}\exp\Big(-\frac{3}{4}(\log X)^{1/10}\Big) + X^{\sigma+o(1)} \ll X(\log X)^{-A}.$$

□

B Proofs of the sieve lemmas

This appendix proves Lemmas 3, 4 and 5 of Section 4 and Lemma 7 of Section 5. In the proofs of Lemmas 4 and 5, we keep every prime of a bilinear estimate at norm at least $\mu^4/2$, and Lemma 3 treats the sifting by the primes of norm below μ^4. We remove the conditions that couple the variables by short intervals. The proof of Lemma 5 iterates Buchstab's identity in stages, one for each length of a chain of primes.

Proof of Lemma 3. For $u \mid n$, let $\mathfrak{n}$ be the number of distinct primes of norm less than $z_0(u)$ that divide n/u. The number of $\mathfrak{d} \mid \mathfrak{P}(z_0(u))$ with $\mathfrak{d} \mid n/u$ and $\mathfrak{o}(\mathfrak{d}) = i$ is $\binom{\mathfrak{n}}{i}$. For $\mathfrak{n} \geqslant 1$,

$$\sum_{i=0}^{M} (-1)^i \binom{\mathfrak{n}}{i} = (-1)^M \binom{\mathfrak{n}-1}{M},$$

and this is at most $\binom{\mathfrak{n}}{M+1}$ in absolute value. Hence

$$S'(\mathcal{E}_u, z_0(u)) = \sum_{\mathfrak{d}} \big(\mathfrak{b}_{\mathfrak{d}}^+(u) - \mathfrak{b}_{\mathfrak{d}}^-(u)\big)|\mathcal{E}_{u\mathfrak{d}}| + \Delta_u, \qquad |\Delta_u| \leqslant \sum_{\mathfrak{d}} \mathfrak{b}_{\mathfrak{d}}^0(u)|\mathcal{E}_{u\mathfrak{d}}|. \tag{33}$$

Multiply by c_u and sum over u. By hypothesis, the first sums on the right of (33) for $\mathcal{A}$ are $2/(\pi W)$ times those for $\mathcal{B}$, up to $\mathcal{O}(XW^{-1}(\log X)^{-A})$. The remainders Δ_u contribute at most

$$\sum_{u,\mathfrak{d}} c_u \mathfrak{b}_{\mathfrak{d}}^0(u)\Big(|\mathcal{A}_{u\mathfrak{d}}| + \frac{2}{\pi W}|\mathcal{B}_{u\mathfrak{d}}|\Big) = \frac{4}{\pi W} \sum_{u,\mathfrak{d}} c_u \mathfrak{b}_{\mathfrak{d}}^0(u)|\mathcal{B}_{u\mathfrak{d}}| + \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big),$$

where the equality holds by hypothesis. By Lemma 18 and Lemma 17(b), with $\sum_{|\varpi|^2<\mu^4} |\varpi|^{-2} \leqslant \log\log X$,

$$\sum_{u,\mathfrak{d}} c_u \mathfrak{b}_{\mathfrak{d}}^0(u)|\mathcal{B}_{u\mathfrak{d}}| \leqslant CX \sum_u c_u |u|^{-2} \frac{(\log\log X)^{M+1}}{(M+1)!} \ll X(\log X)^C \Big(\frac{e\log\log X}{M+1}\Big)^{M+1} \leqslant X(\log X)^C e^{-(M+1)},$$

once $M+1 \geqslant e^2 \log\log X$. Finally $e^{-(M+1)} \leqslant \exp(-(\varepsilon_0/5)(\log X)^{1/5})$ is smaller than every power of $1/\log X$.

It remains to check the last two sentences of the lemma. The indicators $\mathfrak{b}_{\mathfrak{d}}(u)$ vanish unless $|\mathfrak{d}|^2 < \mu^{4(M+1)} \leqslant X^{\varepsilon_0}$. Under either of the two conditions stated there, $\sum_{v=u\mathfrak{d}} c_u \mathfrak{b}_{\mathfrak{d}}(u)$ is non-negative, at most $d(v)^{C_0+1}$, and supported on $|v|^2 \leqslant X^\tau \mu^{-1}$. Under the second condition, each $\mathfrak{d}$ is attached to the factor u_2. The product $u_2\mathfrak{d}$ is the variable u' of Lemma 1(b), and its range is split into dyadic ranges. So Lemma 1 gives the hypothesis. □

The short intervals J_ι and the indices ι_q are those of Section 5. For an index ι with $\xi_{\iota+1} > \mu^4$ we have $\xi_\iota \geqslant \mu^4/2 \geqslant \mu$, so Lemma 17(a) gives

$$\sum_{|\varpi|^2 \in J_\iota} |\varpi|^{-2} \leqslant \frac{2}{H\log(\mu^4/2)}. \tag{34}$$

Let $\mathfrak{s}_\iota$ denote the sum on the left of (34). In sums over ι, the index runs over the ι with $J_\iota \cap [1, 2X] \neq \emptyset$. By Lemma 17(b), $\sum_\iota \mathfrak{s}_\iota \leqslant 2\log\log X$, so that

$$\sum_{\iota:\ \xi_{\iota+1}>\mu^4} \mathfrak{s}_\iota^2 \leqslant \frac{4\log\log X}{H\log(\mu^4/2)}. \tag{35}$$

Proof of Lemma 4. Fix A, let C be such that $\sum_{|n|^2\leqslant X} a_n|n|^{-2} \leqslant (\log X)^C$ (Lemma 18), and take $B = A + C + 3$. Split the support of a into $\mathcal{N}_1 = \{n : l(n) \leqslant \mu^4\}$ and $\mathcal{N}_2 = \{n : l(n) > \mu^4\}$. On $\mathcal{N}_1$, Lemma 3 applies with $c_n = a_n 1_{\mathcal{N}_1}(n)$, the level $z_0 = l$ and $\varepsilon_0 = \varepsilon/2$, since $X^{\tau-\varepsilon} \leqslant X^{\tau-\varepsilon/2}\mu^{-1}$. On $\mathcal{N}_2$, Buchstab's identity (5) gives

$$S'(\mathcal{E}_n, l(n)) = S'(\mathcal{E}_n, \mu^4) - \sum_{\mu^4 \leqslant |q|^2 < l(n)} S'(\mathcal{E}_{nq}, q), \tag{36}$$

and the first term on the right of (36), summed with a_n, has an asymptotic formula by Lemma 3 at the constant level μ^4. It remains to treat

$$\mathcal{G}(\mathcal{E}) = \sum_{n\in\mathcal{N}_2} a_n \sum_{\mu^4\leqslant|q|^2<l(n)} S'(\mathcal{E}_{nq}, q).$$

Two conditions couple n, q and the cofactor: $|q|^2 < l(n)$, and the condition that every prime factor of the cofactor is $\succcurlyeq q$. For $n \in \mathcal{N}_2$ let $\iota(n)$ be the index with $l(n) \in J_{\iota(n)}$. Then

$$1[|q|^2 < l(n)] = 1[\iota_q < \iota(n)] + 1[\iota_q = \iota(n)]\, 1[|q|^2 < l(n)].$$

Moreover, $0 \leqslant S'(\mathcal{E}_{nq}, \xi_{\iota_q}) - S'(\mathcal{E}_{nq}, q)$. This difference is at most the number of prime factors ϖ' of the cofactor with $|\varpi'|^2 \in J_{\iota_q}$, because a prime factor $\varpi' \prec q$ of norm at least ξ_{ι_q} has its norm in J_{ι_q}. With q running over the primes of norm at least μ^4 in J_ι, this gives

$$\Big|\mathcal{G}(\mathcal{E}) - \sum_\iota \mathfrak{M}_\iota(\mathcal{E})\Big| \leqslant \sum_\iota \big(\mathfrak{D}_\iota(\mathcal{E}) + \mathfrak{D}'_\iota(\mathcal{E})\big), \tag{37}$$

where

$$\mathfrak{M}_\iota(\mathcal{E}) = \sum_{\iota(n)>\iota} a_n \sum_q S'(\mathcal{E}_{nq}, \xi_\iota), \quad \mathfrak{D}_\iota(\mathcal{E}) = \sum_{\iota(n)=\iota} a_n \sum_q |\mathcal{E}_{nq}|, \quad \mathfrak{D}'_\iota(\mathcal{E}) = \sum_{\iota(n)\geqslant\iota} a_n \sum_q \sum_{|\varpi'|^2\in J_\iota} |\mathcal{E}_{nq\varpi'}|.$$

Each of these sums has the form

$$\sum_{u,v} b'_u b_v \,\#\{q : |q|^2 \in J_\iota,\ |q|^2 \geqslant \mu^4,\ uvq \in \mathcal{E}\},$$

with $u = n$, with b'_u equal to a_n restricted to $\iota(n) > \iota$, $\iota(n) = \iota$ or $\iota(n) \geqslant \iota$, and with v the cofactor of nq. Its coefficient b_v is the indicator that every prime factor of v has norm at least ξ_ι for $\mathfrak{M}_\iota$, is 1 for $\mathfrak{D}_\iota$, and is the number of primes in J_ι dividing v for $\mathfrak{D}'_\iota$, which is at most $d(v)$. For $\iota(n) \geqslant \iota$ and $|q|^2 \in J_\iota$,

$$|q|^2 < \xi_{\iota+1} \leqslant (1 + 1/H)\, l(n) \leqslant \frac{2C_l X^\tau}{|n|^2},$$

so $|v|^2 \geqslant (X/2)/|nq|^2 \geqslant W/(4C_l)$. In addition, $|n|^2 \geqslant W/C_l$ and $|q|^2 \geqslant \mu^4$. After u and v are split into dyadic ranges, Lemma 2 gives $\mathfrak{M}_\iota$, $\mathfrak{D}_\iota$ and $\mathfrak{D}'_\iota$ asymptotic formulas, uniformly in ι. At most $2H \log X$ indices occur, so Lemma 2 is applied $\mathcal{O}(H(\log X)^3)$ times. With the saving $(\log X)^{-A-B-5}$ in each application, the errors add up to $\mathcal{O}(XW^{-1}(\log X)^{-A})$.

By (37), $\mathcal{G}(\mathcal{A}) - \frac{2}{\pi W}\mathcal{G}(\mathcal{B})$ is $\mathcal{O}(XW^{-1}(\log X)^{-A})$ plus

$$\mathcal{O}\Big(\sum_\iota (\mathfrak{D}_\iota + \mathfrak{D}'_\iota)(\mathcal{A}) + \frac{2}{\pi W}\sum_\iota (\mathfrak{D}_\iota + \mathfrak{D}'_\iota)(\mathcal{B})\Big),$$

and the sums over $\mathcal{A}$ equal $\frac{2}{\pi W}$ times those over $\mathcal{B}$ up to the same error. By Lemma 18, (34) and (35),

$$\sum_\iota \mathfrak{D}_\iota(\mathcal{B}) \leqslant CX \sum_n a_n |n|^{-2} \mathfrak{s}_{\iota(n)}, \qquad \sum_\iota \mathfrak{D}'_\iota(\mathcal{B}) \leqslant CX \sum_n a_n |n|^{-2} \sum_\iota \mathfrak{s}_\iota^2,$$

and both are $\ll X(\log X)^C \log\log X/(H(\log X)^{4/5}) \ll X(\log X)^{-A}$. Hence $\mathcal{G}$ has an asymptotic formula, and so does $\sum_n a_n S'(\mathcal{E}_n, l(n))$. □

Proof of Lemma 5. Fix A, let C be as in the proof of Lemma 4, and take $B = A + C + 4$. The part $\mathcal{N}_1 = \{n : l(n) \leqslant \mu^4\}$ is handled by Lemma 3 with the level l, since $|n|^2 < W \leqslant X^{\tau-2\theta}$. On $\mathcal{N}_2 = \{n : l(n) > \mu^4\}$, (36) holds, and the first term on its right again has an asymptotic formula by Lemma 3. It remains to treat

$$\Sigma_0(\mathcal{E}) = \sum_{n\in\mathcal{N}_2} a_n \sum_{\mu^4\leqslant|q|^2<l(n)} S'(\mathcal{E}_{nq}, q).$$

A *chain of length* $i \geqslant 0$ is a tuple $\mathfrak{c} = (n, p_1, \dots, p_i, q)$ with $n \in \mathcal{N}_2$ and primes $p_1 \succ \cdots \succ p_i \succ q$ of norm at least μ^4, such that the largest of these primes (p_1, or q if $i = 0$) has norm less than $l(n)$, and $|np_1 \cdots p_{i'}|^2 < W$ for $1 \leqslant i' \leqslant i$. Write $N(\mathfrak{c}) = np_1 \cdots p_i q$ and $q(\mathfrak{c}) = q$. Let $\mathcal{K}_i$ be the set of chains of length i, split into the exits $\mathcal{K}_i^{\geqslant}$, with $|N(\mathfrak{c})|^2 \geqslant W$, and the continuations $\mathcal{K}_i^{<}$, with $|N(\mathfrak{c})|^2 < W$, and put

$$\Sigma_i(\mathcal{E}) = \sum_{\mathfrak{c} \in \mathcal{K}_i} a_n S'(\mathcal{E}_{N(\mathfrak{c})}, q(\mathfrak{c})).$$

For $\mathfrak{c} \in \mathcal{K}_i^{<}$, Buchstab's identity (5) gives

$$S'(\mathcal{E}_{N(\mathfrak{c})}, q) = S'(\mathcal{E}_{N(\mathfrak{c})}, \mu^4) - \sum_{q' \prec q,\ |q'|^2 \geqslant \mu^4} S'(\mathcal{E}_{N(\mathfrak{c})q'}, q').$$

The map $(\mathfrak{c}, q') \mapsto (n, p_1, \dots, p_i, q, q')$, on the pairs that occur in these sums, is a bijection onto $\mathcal{K}_{i+1}$. Hence $\Sigma_i = \mathfrak{E}_i + \mathfrak{F}_i - \Sigma_{i+1}$, where

$$\mathfrak{E}_i(\mathcal{E}) = \sum_{\mathfrak{c} \in \mathcal{K}_i^{\geqslant}} a_n S'(\mathcal{E}_{N(\mathfrak{c})}, q(\mathfrak{c})), \qquad \mathfrak{F}_i(\mathcal{E}) = \sum_{\mathfrak{c} \in \mathcal{K}_i^{<}} a_n S'(\mathcal{E}_{N(\mathfrak{c})}, \mu^4).$$

A chain of length i has $\mu^{4i} \leqslant |p_1 \cdots p_i|^2 < W$, so $\mathcal{K}_i$ is empty for $i > i_{\max} = \lfloor \sigma(\log X)^{1/5}/4 \rfloor$. Hence, for $\mathcal{E} = \mathcal{A}$ and for $\mathcal{E} = \mathcal{B}$,

$$\Sigma_0 = \sum_{i=0}^{i_{\max}} (-1)^i (\mathfrak{E}_i + \mathfrak{F}_i). \tag{38}$$

The primes of a chain are distinct and their order is forced, so a chain is determined by its product and the set of its primes. The number of chains with a given product N is therefore at most $d(N)$, and every coefficient defined from chains in this proof is at most $d(N)^{C_0+1}$, uniformly in i. The sum $\mathfrak{F}_i$ is $\sum_N c_i(N) S'(\mathcal{E}_N, \mu^4)$, where $c_i(N)$ is the sum of a_n over the chains $\mathfrak{c} \in \mathcal{K}_i^{<}$ with $N(\mathfrak{c}) = N$. This $c_i(N)$ is supported on $|N|^2 < W$, so Lemma 3 gives $\mathfrak{F}_i$ an asymptotic formula, uniformly in i.

For the exits, let $j_{\max} = \lceil \sigma/\theta \rceil$ and $C_E = 2C_l^{j_{\max}+2}$. We claim that every $\mathfrak{c} \in \mathcal{K}_i^{\geqslant}$ satisfies

$$W \leqslant |N(\mathfrak{c})|^2 \leqslant C_E X^{\tau-\theta/2}, \qquad |N(\mathfrak{c})|^2 |q(\mathfrak{c})|^2 \leqslant C_E X^{\tau}. \tag{39}$$

Write the primes of $\mathfrak{c}$ as $p_1 \succ \cdots \succ p_k$ with $p_k = q$ and $k = i + 1$, and put $N' = N(\mathfrak{c})/q$, so that $|N'|^2 < W \leqslant |N(\mathfrak{c})|^2$. Let $j = j(|n|^2)$. Then $0 \leqslant j < \sigma/\theta \leqslant j_{\max}$. Since $|n|^2 \geqslant WX^{-(j+1)\theta}$, we have $z(|n|^2) \leqslant X^{(j+3)\theta/(j+2)} \leqslant X^{3\theta/2}$, and

$$|n|^2 |p_1|^{2(j+2)} < |n|^2 l(n)^{j+2} \leqslant C_l^{j+2} X^{\tau}.$$

If $|q|^2 \leqslant X^{\theta}$, then $|N(\mathfrak{c})|^2 |q|^2 = |N'|^2 |q|^4 < WX^{2\theta} = X^{\tau}$. If $|q|^2 > X^{\theta}$, then every prime p_h of $\mathfrak{c}$ has norm greater than X^{θ}, so

$$|n|^2 X^{(k-1)\theta} \leqslant |N'|^2 < W \leqslant |n|^2 X^{(j+1)\theta}.$$

This gives $k \leqslant j + 1$, and then

$$|N(\mathfrak{c})|^2 |q|^2 \leqslant |n|^2 |p_1|^{2(k+1)} \leqslant |n|^2 |p_1|^{2(j+2)} \leqslant C_l^{j+2} X^{\tau}.$$

It remains to bound $|N(\mathfrak{c})|^2$. If $k \geqslant 2$, then

$$|N(\mathfrak{c})|^2 < W|q|^2 \leqslant W|p_1|^2 < C_l W X^{3\theta/2} = C_l X^{\tau-\theta/2}.$$

If $k = 1$, then

$$|N(\mathfrak{c})|^2 = |n|^2 |q|^2 < C_l |n|^2 z(|n|^2) = C_l |n|^{2(j+1)/(j+2)} X^{\tau/(j+2)} \leqslant C_l X^{(\sigma(j+1)+\tau)/(j+2)} \leqslant C_l X^{1/2}.$$

Now split the last prime into short intervals. For $\mathfrak{c} \in \mathcal{K}_i^{\geqslant}$ with $\iota_{q(\mathfrak{c})} = \iota$, the difference $S'(\mathcal{E}_{N(\mathfrak{c})}, \xi_\iota) - S'(\mathcal{E}_{N(\mathfrak{c})}, q(\mathfrak{c}))$ lies between 0 and the number of prime factors of the cofactor with norm in J_ι. Hence

$$\Big| \mathfrak{E}_i - \sum_{\iota} \sum_{n''} a''_{i,\iota}(n'') S'(\mathcal{E}_{n''}, \xi_\iota) \Big| \leqslant \mathfrak{E}'_i = \sum_{\iota} \sum_{n''} a''_{i,\iota}(n'') \sum_{|\varpi'|^2 \in J_\iota} |\mathcal{E}_{n''\varpi'}|, \tag{40}$$

where $a''_{i,\iota}(n'')$ is the sum of a_n over the chains $\mathfrak{c} \in \mathcal{K}_i^{\geqslant}$ with $N(\mathfrak{c}) = n''$ and $\iota_{q(\mathfrak{c})} = \iota$. By (39), on the support of $a''_{i,\iota}$ we have $W \leqslant |n''|^2 \leqslant C_E X^{\tau-\theta/2}$ and $\xi_\iota \leqslant |q|^2 \leqslant C_E X^\tau/|n''|^2$. So Lemma 4, with the constant level ξ_ι, with $\theta/4$ in place of ε and with C_E in place of C_l, gives every main term in (40) an asymptotic formula. The sum $\mathfrak{E}'_i$ is of the form (15) with $u = n''$, with the prime ϖ' and with v the cofactor. The norm of ϖ' lies in J_ι and is at least $\mu^4/2$, and $|v|^2 \geqslant (X/2)/(|n''|^2\xi_{\iota+1}) \geqslant W/(3C_E)$. So Lemma 2 gives it an asymptotic formula.

The *stage* i consists of the sums $\mathfrak{E}_i$ and $\mathfrak{F}_i$. Each stage has at most $2H\log X$ indices ι, at most $\log X$ dyadic ranges of n'' and at most $\log X$ dyadic ranges of v. So (38) and (40) express Σ_0 through at most $2H(\log X)^2$ applications of Lemma 4, at most $2H(\log X)^3$ of Lemma 2 and 1 of Lemma 3 per stage, over at most $(\log X)^{1/5}$ stages. These are $\mathcal{O}(H(\log X)^{16/5})$ sums in all. With the saving $(\log X)^{-A-B-5}$ in each, their errors add up to $\mathcal{O}(XW^{-1}(\log X)^{-A})$.

It remains to bound the remainders $\mathfrak{E}'_i$. By Lemma 18,

$$\sum_i \mathfrak{E}'_i(\mathcal{B}) \leqslant CX \sum_n a_n |n|^{-2} \sum_{i\geqslant 0} \sum_{p_1 \succ \cdots \succ p_i} \prod_h |p_h|^{-2} \sum_\iota \mathfrak{s}_\iota^2,$$

where the factor $\mathfrak{s}_\iota^2$ comes from the primes q and ϖ', which both lie in J_ι. The sum over decreasing chains of i distinct primes of norm at most X is at most $(2\log\log X)^i/i!$, so the sum over i is at most $(\log X)^2$. With (35),

$$\sum_i \mathfrak{E}'_i(\mathcal{B}) \ll \frac{X(\log X)^{C+2}\log\log X}{H(\log X)^{4/5}} \ll X(\log X)^{-A},$$

and the same bound holds for $\sum_i \mathfrak{E}'_i(\mathcal{A})$ up to the factor $2/(\pi W)$. Summing (38) term by term gives

$$\Sigma_0(\mathcal{A}) - \frac{2}{\pi W}\Sigma_0(\mathcal{B}) = \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big).$$

□

Proof of Lemma 7. Fix A and take $B = A+1$. For a tuple ϖ, let $\iota(\varpi) = (\iota_{\varpi_1}, \dots, \iota_{\varpi_k})$ be its index vector. For a constraint c, let $g_c(\iota) = 1$ if ι satisfies c and $h_c(\iota) = 1$ if ι straddles c, and let both be 0 otherwise. An index vector that neither satisfies nor straddles c has no tuple satisfying c, so $1_c(\varpi) = g_c(\iota(\varpi)) + 1_c(\varpi)h_c(\iota(\varpi))$, and $g_c h_c = 0$. Let $\Xi_\iota(t)$ be the indicator that every prime factor of t has norm at least y_ι. For $\iota = \iota(\varpi)$ we have $\Xi(\varpi, t) \leqslant \Xi_\iota(t)$, and the difference is at most the number of prime factors of t with norm in J_κ for some $\kappa \in \mathcal{W}(\iota)$. Indeed, in type (S), a prime factor $\varpi' \prec \varpi_{i_s}$ of t with $|\varpi'|^2 \geqslant \xi_{\iota_{i_s}}$ has its norm in $J_{\iota_{i_s}}$. In type (M), a prime factor ϖ' of t with $\inf_{\mathcal{T}(\iota)} m_s \leqslant |\varpi'|^2 < m_s(\varpi)$ has its norm in J_κ for some $\kappa \in \mathcal{W}(\iota)$. In types (C) and (F), $\Xi_\iota = \Xi$. All these indicators take values in $\{0,1\}$, so for $\iota = \iota(\varpi)$,

$$\Big|\prod_c 1_c(\varpi)\,\Xi(\varpi,t) - \prod_c g_c(\iota)\,\Xi_\iota(t)\Big| \leqslant \sum_c h_c(\iota) + \prod_c g_c(\iota)\,\big(\Xi_\iota(t) - \Xi(\varpi,t)\big). \tag{41}$$

Let $\mathfrak{I}_0$ be the set of index vectors that satisfy all constraints, and $\mathfrak{I}_c$ the set of those that straddle c. Multiply (41) by $1[\varpi_1\cdots\varpi_k t \in \mathcal{E}]\prod_i 1[|\varpi_i|^2 \in \mathcal{I}_i]$ and sum. This gives

$$\Big|\Sigma_{\mathcal{C}}(\mathcal{E}) - \sum_{\iota\in\mathfrak{I}_0} \mathcal{P}_\iota(\mathcal{E}; y_\iota)\Big| \leqslant E^+(\mathcal{E}), \tag{42}$$

where

$$E^+(\mathcal{E}) = \sum_c \sum_{\iota\in\mathfrak{I}_c} \mathcal{P}_\iota(\mathcal{E};1) + \sum_{\iota\in\mathfrak{I}_0}\sum_{\kappa\in\mathcal{W}(\iota)} \mathcal{P}'_{\iota,\kappa}(\mathcal{E}).$$

At most $(2H\log X)^k$ index vectors have $\mathcal{T}(\iota) \neq \emptyset$. By (42) for $\mathcal{A}$ and for $\mathcal{B}$, and by the hypotheses with $A' = A + (k+1)(A+3)$ in place of A, we obtain

$$\Sigma_{\mathcal{C}}(\mathcal{A}) - \frac{2}{\pi W}\Sigma_{\mathcal{C}}(\mathcal{B}) = \mathcal{O}\Big(E^+(\mathcal{A}) + \frac{2}{\pi W}E^+(\mathcal{B})\Big) + \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big),$$
$$E^+(\mathcal{A}) = \frac{2}{\pi W}E^+(\mathcal{B}) + \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big).$$

It therefore suffices to show

$$E^+(\mathcal{B}) \ll \frac{X(\log\log X)^k}{H(\log X)^{4/5}}. \tag{43}$$

By Lemma 18(a), a tuple ϖ contributes at most $CX/|\varpi_1\cdots\varpi_k|^2$ to $\mathcal{P}_\iota(\mathcal{B};1)$, and a tuple (ϖ,ϖ') at most $CX/|\varpi_1\cdots\varpi_k\varpi'|^2$ to $\mathcal{P}'_{\iota,\kappa}(\mathcal{B})$. If ι straddles c, every tuple of $\mathcal{T}(\iota)$ has $m_c(\varpi)$ within the factor $(1+1/H)^{2k}$ of Z_c. Choose i_1 with $e_{c,i_1}\neq 0$. For fixed ϖ_i, $i\neq i_1$, the norm $|\varpi_{i_1}|^2$ then lies in an interval $[Y, Y(1+1/H)^{4k})$ with $Y\geqslant\mu^4$, and Lemma 17(a) bounds the sum of $|\varpi_{i_1}|^{-2}$ over it by $8k/(H\log\mu^4)$. Each other variable contributes at most $\sum_{|\varpi|^2\leqslant X}|\varpi|^{-2}\leqslant 2\log\log X$. This bounds the sums over $\mathfrak{I}_c$ in $E^+(\mathcal{B})$ by the right side of (43). In the sum over $\mathfrak{I}_0$, for fixed ι, the norm of ϖ' runs over an interval $[Y', Y'(1+1/H)^{2k+2})$ with $Y'\geqslant\mu^4/2$, so that

$$\sum_{\varpi'}|\varpi'|^{-2}\leqslant\frac{4k+4}{H\log(\mu^4/2)}.$$

The remaining sums over $\mathcal{T}(\iota)$ add up, over all ι, to at most $(2\log\log X)^k$. This bounds the sum over $\mathfrak{I}_0$ by the right side of (43).

The bound (43) also covers the terms of $\Sigma_{\mathcal{C}}$ in which two primes have equal norm, and those in which a prime factor of t equals ϖ_{i_s}. Two primes of equal norm lie in the same short interval, so the index vector of such a tuple straddles the order constraint between them. A prime factor of t equal to ϖ_{i_s} has its norm in J_κ with $\kappa=\iota_{i_s}\in\mathcal{W}(\iota)$. On the side of $\mathcal{A}$, the straddling pieces and the window pieces are controlled through their own asymptotic formulas, hypotheses (b) and (c). □

C Evaluation of discarded sums by integrals

This appendix proves Lemma 15, which the proof of Lemma 16 uses to bound the discarded sums over $\mathcal{B}$.

Proof of Lemma 15. We first count tuples with a free prime. Let $K'\geqslant 1$, and let $\mathcal{R}'\subset\mathbb{R}^{K'}$ be bounded, with boundary of measure 0, and with $y_i\geqslant c_1$ and $\sum_i y_i\leqslant 1-c_1$ on $\mathcal{R}'$. We claim

$$\#\big\{(\varpi_0,\dots,\varpi_{K'}) : y(\varpi_1,\dots,\varpi_{K'})\in\mathcal{R}',\ \varpi_0\cdots\varpi_{K'}\in\mathcal{B}\big\}=\frac{X}{2\log X}\Big(\int_{\mathcal{R}'}\frac{dy}{y_1\cdots y_{K'}(1-\sum_i y_i)}+o(1)\Big). \tag{44}$$

For fixed $\varpi_1,\dots,\varpi_{K'}$ with $Y=\prod_i|\varpi_i|^2\leqslant X^{1-c_1}$, Lemma 17(d) gives the number of ϖ_0 as

$$\pi_G(X/Y)-\pi_G(X/(2Y))=(1+o(1))\frac{X}{2Y(1-\sum_i y_i)\log X}.$$

So the left side of (44) is

$$(1+o(1))\frac{X}{2\log X}\sum_{\varpi_1,\dots,\varpi_{K'}}F(y(\varpi))\prod_i|\varpi_i|^{-2},\qquad F=\frac{1_{\mathcal{R}'}}{1-\sum_i y_i}.$$

Let $\mathfrak{m}_X$ be the measure on $[c_1,1]$ that gives mass $|\varpi|^{-2}$ to the point $\log|\varpi|^2/\log X$ for each prime ϖ with $|\varpi|^2\geqslant X^{c_1}$. By Lemma 17(c), the $\mathfrak{m}_X^{\otimes K'}$-measure of every box $\prod_i[a_i,b_i)$ tends to $\prod_i\log(b_i/a_i)$, the mass of the box under $\prod_i dy_i/y_i$. The function F is bounded and continuous outside the null set $\partial\mathcal{R}'$, hence Riemann integrable: for every $\eta'>0$ there are step functions $F_-\leqslant F\leqslant F_+$ on a finite grid of boxes with $\int(F_+-F_-)\prod_i dy_i/y_i<\eta'$. This gives (44).

Now consider the left side of (27). For each tuple ϖ in its range, $S(\mathcal{B}_{\varpi_1\cdots\varpi_K},\varpi_K)$ counts the cofactors $t=\varpi'_1\cdots\varpi'_h$ with $\varpi_K\preccurlyeq\varpi'_1\preccurlyeq\cdots\preccurlyeq\varpi'_h$ and $\varpi_1\cdots\varpi_K t\in\mathcal{B}$. Here $t\neq 1$ because $\sum_i y_i\leqslant 1-c_1$, and $h\leqslant 1/c_1$. The tuples with a repeated prime, or with $\varpi'_1=\varpi_K$, number $\ll\sum_{|\varpi'|^2\geqslant X^{c_1}}X|\varpi'|^{-4}\ll X^{1-c_1}$. The other tuples have distinct norms apart from a set of the same size, and they are $1/h!$ of the ordered tuples. Apply (44) with $K'=K+h-1$ and the free prime $\varpi_0=\varpi'_h$ to the region

$$\mathcal{R}'_h=\Big\{(y,x_1,\dots,x_{h-1}) : y\in\mathcal{R},\ x_i\geqslant y_K,\ 1-\sum_i y_i-\sum_i x_i\geqslant y_K\Big\},$$

where $x_i = \log|\varpi_i'|^2/\log X$ is the exponent of ϖ_i'. The condition $1-\sum_i y_i - \sum_i x_i \geqslant y_K$ of $\mathcal{R}_h'$ is the condition on ϖ_h', up to $\mathcal{O}(1/\log X)$. It is handled by placing $\mathcal{R}_h'$ between its shifts by $\pm\eta''$ and letting $\eta''\to 0$. The integrand $\prod_{i\leqslant h} 1/x_i$, with $x_h = w - \sum_{i<h} x_i$ and $w = 1-\sum_i y_i$, is symmetric in the h coordinates, so $1/h!$ of the integral over $\mathcal{R}_h'$ is the integral over the part with $x_1 \leqslant \ldots \leqslant x_h$. Hence the left side of (27) is

$$\frac{X}{2\log X}\Big(\int_{\mathcal{R}} g\Big(1-\sum_i y_i;\, y_K\Big)\frac{dy}{y_1\cdots y_K} + o(1)\Big),$$

where

$$g(w;b) = \sum_{h\geqslant 1}\int_{b\leqslant x_1\leqslant\ldots\leqslant x_{h-1}\leqslant w-\sum_{i<h}x_i}\prod_{i<h}\frac{dx_i}{x_i}\cdot\Big(w-\sum_{i<h}x_i\Big)^{-1}.$$

Splitting off the smallest coordinate $x = x_1 \in [b, w/2]$ when $h \geqslant 2$ gives

$$g(w;b) = \frac{1}{w} + \int_b^{w/2} g(w-x;x)\,\frac{dx}{x} \quad (w\geqslant b), \qquad g(w;b) = 0 \quad (w<b). \tag{45}$$

The function $\omega(w/b)/b$ satisfies (45), since the substitution $x = w/(u+1)$ and the relation that defines ω below (13) give

$$\int_b^{w/2}\omega\Big(\frac{w-x}{x}\Big)\frac{dx}{x^2} = \frac{1}{w}\int_1^{w/b-1}\omega(u)\,du, \qquad \frac{1}{w}\Big(1+\int_1^{w/b-1}\omega(u)\,du\Big) = \frac{1}{b}\,\omega\Big(\frac{w}{b}\Big).$$

Both functions vanish for $w<b$, and (45) determines them by induction on $\lceil w/b\rceil$, since $(w-x)/x \leqslant w/b-1$. So $g(w;b) = \omega(w/b)/b$, and (27) follows from (26). □

D Inequalities used in the decomposition

Table 4 lists the inequalities between the parameters (3) that the decomposition uses, directly or through the lemmas of Sections 4–6. The exponent $\chi = (\nu+2\theta)/2$ in the table is that of Section 7. The table writes each inequality as a quantity that must be positive. These quantities are linear in θ and non-negative at $\theta = 1/9$ and $\theta = 1/8$, hence positive for $1/9 < \theta < 1/8$, the range of Section 7.

E Departures from the argument of Harman and Lewis

Table 5 lists the departures of this paper from the argument of Harman and Lewis [7]. The paragraphs and remarks after the table give the details, in the order of the table.

Remark 24 *In their (27) and (28), Harman and Lewis print the conditions $uv \in \mathcal{A}$ and $uvt \in \mathcal{A}$, here written in the notation of Lemma 23. The proof of their Lemma 6 selects the norm range by the factor $(X/|u|^2)^s(1-2^{-s})$ in a Perron integral of $L(s,\lambda^m)$, with no condition on the argument, and the proof of their Lemma 7 proceeds in the same way. So the correct conditions are $uv \in \mathcal{B}$ and $uvt \in \mathcal{B}$, which is also the form needed in Lemma 20. Their stated saving is $X\eta$, for their small parameter η. It holds for every $\eta = (\log X)^{-A}$, since the proofs save $\exp(-c(\log X)^{1/10})$ or a power of μ. The hypothesis $\theta > 1/14$ of their Lemma 7 holds throughout this paper.*

Remark 25 *Harman and Lewis [7, Lemma 5] state the bilinear estimate for $U, V \geqslant W$ and $UV \leqslant X\mu^{-1}$, with the prime ϖ running over all norms up to $4X/(UV)$. Their proof applies Lemma 22 with $|t| \leqslant X$ and $S = 4X/(UV)$. As stated, Lemma 22 then requires*

$$S \geqslant \exp\big(((1+\sigma)\log X + A\log\log X)^{4/5}\big),$$

and this lower bound is about $\mu^{1.3}$. This requirement fails when UV is close to $X\mu^{-1}$. Lemma 2 avoids this range: its prime has norm at least $\mu^4/2$. Every use of the bilinear estimate in this paper goes through Lemma 2, and primes of norm below μ^4 are handled by the fundamental lemma, Lemma 3.

Quantity	Inequality and use	Value at $\theta = 0.1129$
$\theta - 1/10$	$3\sigma/2 < \tau$; Lemma 12	0.0129000
$1/8 - \theta$	range of θ	0.0121000
$(1 - 8\theta)/3$	$\nu + \rho < 1$; Lemma 9	0.0322667
$1/6 - \theta$	Lemma 2	0.0537667
$2\theta - 1/6$	$\sigma - \theta < 1/3$; $j = 0$ in Range 2	0.0591333
$3\theta/2 - 1/12$	$\sigma < (1/3 + \tau)/2$; $PQ \geqslant W$ in Range 2 and on R	0.0860167
$(3 - 22\theta)/12$	$2\zeta - 1/3 < \nu$; $PQ^2 \leqslant X^\nu$ in Range 2	0.0430167
$(1 + 6\theta)/12$	$\zeta - 1/3 < \rho$; $Q \leqslant X^\rho$ in Range 2	0.1397833
$(1 - 2\theta)/6$	$\zeta < \tau$; $PQ \leqslant X^{\tau-\varepsilon}$ where $\alpha + \beta \leqslant \zeta$	0.1290333
$(14\theta - 1)/6$	$\sigma < \zeta$; $P \leqslant X^\zeta$ on R	0.0967667
$(5 - 34\theta)/24$	$\zeta - \sigma < \rho/2$; $\mathsf{R} \neq \emptyset$	0.0483917
$(26\theta - 1)/24$	$\sigma + \rho/2 < \tau$; $PQ \leqslant X^{\tau-\varepsilon}$ on R	0.0806417
$(46\theta - 5)/12$	$\sigma - \theta < \rho$; $j = 0$ in Range 3, $Q \leqslant X^\rho$ in Range 5	0.0161167
$(58\theta - 5)/24$	$\sigma < \chi$; $PQ \geqslant W$ in Range 3	0.0645083
$(1 - 2\theta)/24$	$\chi < \zeta$; Lemma 9 in Range 1	0.0322583
$(11 - 82\theta)/36$	$\nu/3 < \sigma - \theta$; $\mathsf{D}_8 \neq \emptyset$	0.0483944
$(46\theta - 5)/24$	$(\nu + \sigma - \theta)/2 < \zeta$; $PQ \leqslant X^\zeta$ in Range 5	0.0080583
$5\theta/3 - 1/6$	$\sigma < 2\tau/3$; $PQ \geqslant W$ on E_9	0.0215000
$5\theta/2 - 1/4$	$\sigma - \theta < \tau/2$; E_{10} lies in Range 5	0.0322500
$17\theta/5 - 3/10$	$\sigma - 2\theta < 2\tau/5$; $\lvert pq\rvert^2 \geqslant WX^{-2\theta}$ in Range 6	0.0838600
$21\theta/5 - 2/5$	$\sigma - 3\theta < \tau/5$; $j(X^\alpha) \leqslant 2$ in Range 6	0.0741800

Table 4: Inequalities between the parameters (3), written as quantities that must be positive, with the place where each is used and its value at $\theta = 0.1129$.

Harman and Lewis invoke "a fundamental lemma" in the proofs of their Lemmas 8, 11 and 12, without a statement. Lemma 3 states the form used in this paper, at levels up to μ^4, and Appendix B proves it by Brun's pure sieve.

Lemmas 4 and 5 and Corollary 6 prove Lemmas 8, 9 and 10 of Harman and Lewis [7], which give asymptotic formulas for $\sum_n a_n S(\mathcal{E}_n, l(n))$. The proofs of Harman and Lewis apply the bilinear estimate with a prime of norm down to μ, below the range of Lemma 2. For the prime q of the first Buchstab step, they leave the conditions $|q|^2 < l(n)$ and "every prime factor of the cofactor is $\geqslant q$" to Perron's formula. Their Lemma 9 iterates Buchstab's identity up to $(\log X)^{1/5}$ times. The proof of Lemma 5 organizes this iteration into stages, with coefficients bounded uniformly in the stage.

Lemma 8 of Harman and Lewis [7] is the case of Lemma 4 with $n \sim Y$ and $l(n) = R|n|^k$, where $-2 \leqslant k \leqslant 2$ and $\mu \leqslant l(n)$.

Lemma 9 of Harman and Lewis [7], for $n \sim Y$ with $X^\theta \ll Y \ll W$ and $l(n) = R|n|^k \ll z(Y)$, follows from Lemmas 5 and 4. For the n with $|n|^2 < W$, Lemma 5 applies, since then $z(|n|^2) \geqslant z(Y)/2$. For the n with $|n|^2 \geqslant W$, Lemma 4 applies, since $z(Y) \leqslant X^\tau/(2Y) \leqslant X^\tau/|n|^2$. Lemma 10 of Harman and Lewis [7] is Corollary 6.

Harman and Lewis remove the conditions that couple the variables of their sums by "Perron's formula ... to disentangle" and by "removing the interdependence between s and t" [7, pp. 129–130]. Lemma 7 removes them by short intervals.

In their Lemma 11, Harman and Lewis assume $W \leqslant P \ll X^\chi$ with $\chi = (34\theta+7)/24$, and $\max(X^\tau/P, PX^{-2\theta}) \ll Q \ll X^\nu/(P\mu)$. Since $\chi < \zeta$, their hypotheses imply those of Lemma 9 on boxes at distance at least ε from the boundary of their range. Their lower bound X^τ/P for Q is not used.

The proof of Lemma 10 applies without the constraint $r \prec q$, and, when $P \geqslant \inf \mathcal{Q}$, without the constraint $q \prec p$, since dropping a constraint changes none of its checks. Lemma 12 of Harman and Lewis [7] is case (a) of Lemma 10 with $\mathcal{Q} = [Q, 2Q)$ and $\mathcal{I} = [R, 2R)$ for $R \geqslant X^\tau/(PQ)$, without the constraints. Their hypothesis $R \geqslant X^\theta$ is not used in the proof. In their Lemma 13, where Lemma 12 is applied, $R \geqslant X^\tau/(PQ) \geqslant X^{\tau-\zeta} > X^\theta$ holds automatically. In the decomposition of Section 7, the inequality $R \geqslant X^\theta$ does not always hold.

Item of Harman and Lewis [7]	Departure	Treated in
(27), (28)	conditions $uv \in \mathcal{A}$, $uvt \in \mathcal{A}$ read as $\mathcal{B}$	Remark 24
Lemma 5 and (30)	Lemma 22 applied below its range	Remark 25
"a fundamental lemma" in Lemmas 8, 11, 12	not stated	Lemma 3
Lemmas 8–10	proofs sketched	Lemmas 4, 5, Corollary 6
Lemmas 11–13, (30)	conditions coupling variables	Lemma 7
Lemma 11	lower bound X^τ/P for Q not needed	Lemma 9
Lemma 12	hypothesis $R \geqslant X^\theta$ not needed	Lemma 10
Lemma 13	levels not of the form $R\lvert n\rvert^k$; case (b) added	Lemma 11
case 6 of §5	step used without a statement	Lemma 12
(37), second term	level above the prime for small $\lvert p\rvert^2$	Lemma 13, Remark 26
§5	decomposition extended to $\theta < 3/26$	Section 7
value of χ, regions of φ_2 and φ_{10}	misprints	Remark 27
cuts in φ_3, φ_7, φ_9	a grouping of G; a cut implied by the limits of δ	Section 8; (23) and (13)
(17)	factor $1 + \mathcal{O}(\eta)$	Lemma 15

Table 5: Departures from the argument of Harman and Lewis [7], with the numbering of lemmas and displays of Harman and Lewis in the first column, and the place where each is treated in this paper. The symbols η, χ and φ_k are those of Harman and Lewis [7].

Lemma 13 of Harman and Lewis [7] is case (a) of Lemma 11. Their condition "$Q^2P < X^\tau$" is the case $S_\geqslant = 0$. Their hypotheses imply $Q \leqslant X^{\rho-\varepsilon}$ for small ε, because $Q^2 \leqslant 2PQ \leqslant 2X^\zeta$ and $\rho - \zeta/2 = (14\theta - 1)/12 > 0$. Their proof applies Lemmas 8 and 12 to the levels $\lvert q\rvert^2$ and $X^\tau/(PQ)$, which are not of the form $R\lvert n\rvert^k$ of their Lemma 8. Lemmas 4 and 7 of this paper allow these levels. Case (b) of Lemma 11 has no counterpart in their Lemma 13. Section 3 describes its use on R.

For $j = 0$, Lemma 12 is Lemma 14 and (36) of Harman and Lewis [7]. Their condition $PQ^3 < X^\tau$ is the case $S_\geqslant = 0$. For $j = 1$ it is the step that Harman and Lewis use without a statement in case 6 of their §5.

Remark 26 *Harman and Lewis [7, (37)] write the second term as the sum of $S(\mathcal{E}_p, z(\lvert p\rvert^2))$ over all p with $X^\theta \leqslant \lvert p\rvert^2 < X^{1/2}$. For $p \in \Pi_2$ the function $S(\mathcal{E}_p, z(\lvert p\rvert^2))$ sifts p itself and vanishes, while the term $S(\mathcal{E}_p, p)$ of the first Buchstab step remains. Lemma 13 gives the correct term.*

For $\theta < 3/26$, the interval $[\tau/4, \sigma - 2\theta)$ is not empty, and its primes have $j(X^\alpha) = 2$. The decomposition of Harman and Lewis [7, §5] does not treat these primes. Section 7 places their pairs in Ranges 5 and 6 and their discarded third sums in E_{10} and E_{11}.

Remark 27 *The regions of Section 7 are derived from the lemmas. They correct two regions printed by Harman and Lewis. Harman and Lewis [7, p. 132] print the lower limit $\zeta - \alpha$ for β in φ_2. For $\alpha > (1+10\theta)/6$, this limit lies below $\beta_0(\alpha) = (\tau - \alpha)/2$, and the strip between these limits contains no pair of $\sum^*$. They print the upper limit $1/2 - 2\theta - \alpha$ for β in φ_{10}, which makes the region empty for every $\alpha \geqslant \tau/4$. The condition $\lvert pq\rvert^2 < W$ of Lemma 12 with $j = 0$ gives the upper limit $\sigma - \alpha$. They also print $\chi = 0.46205$ at $\theta = 0.119$, while $(34\theta + 7)/24 = 0.46025$.*

The regions of Harman and Lewis for φ_3, φ_7 and φ_9 carry further cuts. Harman and Lewis [7, p. 133] use a grouping of the kind that defines G and L only in their φ_7. There, the condition $\delta > 1 - \alpha - \gamma - \sigma$ when $\alpha + \gamma \geqslant \sigma$ excludes from the region of φ_7 the subset of G given by $\mathcal{S} = \{p, r\}$. Their cut $\gamma \leqslant \alpha + \beta - 2\theta$ in φ_3 and φ_9 is implied by the limits of δ. On E_3 and E_9, $\delta \geqslant \tau - \alpha - \beta$ by (23), and the integrand of $\Phi(\mathsf{F}')$ in (13) vanishes unless $\delta \leqslant (1 - \alpha - \beta - \gamma)/2$. These bounds are compatible only if $\gamma \leqslant 1 - 2\tau + \alpha + \beta = \alpha + \beta - 2\theta$.

For $K = 2$, Lemma 15 is Harman and Lewis [7, (17)], with the term $o(1)$ of (27) in place of their factor $1 + \mathcal{O}(\eta)$, where $\eta = (\log X)^{-A}$ as in Remark 24. For $K = 4$ it is the form they use for φ_3, φ_5, φ_7, φ_9, φ_{10} and φ_{11}. The passage from the prime number theorem to the integral already loses $\mathcal{O}(1/\log X)$ through Lemma 17(c), so the factor $1 + \mathcal{O}(\eta)$ is not available in general.

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